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The Physical, Emotional, and Autonomous Anthropomorphism of Service Chatbots

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ABSTRACT

Prior research on chatbot anthropomorphism has often treated it as a unidimensional construct, producing inconsistent findings regarding how human-like design cues shape user responses. This study conceptualizes anthropomorphism as a multidimensional higher-order perception comprising three distinct forms of anthropomorphic attribution (physical, emotional, and autonomous) and examines how each dimension independently and interactively influences customer satisfaction with service chatbots. Drawing on the stereotype content model and expectancy violation theory, the study proposes that different anthropomorphism dimensions activate distinct evaluative pathways: physical and emotional cues primarily signal warmth, whereas autonomous cues primarily signal competence. A mixed-method approach was employed across four studies, combining LLM-based text analysis with experimental designs. Study 1 identifies the three dimensions from consumers' own descriptions of chatbot human-likeness. Study 2 demonstrates that autonomous anthropomorphism uniquely drives satisfaction through perceived competence, whereas emotional anthropomorphism enhances perceived warmth, which does not translate into higher satisfaction in utilitarian service contexts, and physical anthropomorphism shows no significant effect on satisfaction. Study 3 reveals a counterintuitive interaction: the positive effect of autonomous anthropomorphism on satisfaction is attenuated when emotional anthropomorphism is high, suggesting that warmth-signaling cues can undermine competence-driven benefits in utilitarian service contexts. Study 4 replicates the interaction found in Study 3, and confirms that the attenuation happens in the translation of autonomous cues into competence perceptions and it is this diluted competence perception, not any change in warmth, that carried the interaction into satisfaction. By disentangling anthropomorphism into discrete dimensions and empirically testing their interplay, this research provides a structured lens through which previously mixed findings can be better understood. The findings challenge managers to prioritize autonomous capabilities over superficial human-like traits when designing AI-driven services, particularly in efficiency-focused industries such as banking and e-commerce.

1 | Introduction

Service chatbots, systems that interact with users through natural language to handle tasks such as booking, information retrieval, and complaint resolution, have become a dominant channel for customer service across banking, e-commerce, hospitality, and healthcare (Blut et al. 2021).

The scale of this shift is substantial: the global chatbot market is projected to exceed \$27 billion by 2030 (Grand View Research 2024), with customer service representing the largest application segment. As organizations increasingly rely on automated conversational agents to manage customer interactions, understanding how chatbot design

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shapes user responses has become an important question for both researchers and practitioners.

With the rapid advancement of large language models (e.g., ChatGPT, Claude, Gemini), these chatbots are increasingly capable of mimicking human-like communication, enabling organizations to design service agents that look, sound, and behave more like humans than ever before. A central design question has therefore emerged: how should chatbot anthropomorphism, that is, the attribution of human-like characteristics to non-human agents (Epley et al. 2007), be designed to effectively enhance customer experience?

A growing body of research has examined the effects of anthropomorphic design cues on user responses to chatbots and service robots. However, this literature has produced notably inconsistent findings. For instance, Araujo (2018) found that anthropomorphic cues such as human-like language increased perceived social presence but did not significantly improve attitudes toward the company. Yuan and Dennis (2019) demonstrated that visual anthropomorphism increased willingness to pay in e-commerce, yet auditory anthropomorphism did not produce the same effect. In a virtual reality context, Stein and Ohler (2017) reported that attributing autonomous empathy to a virtual agent increased perceived eeriness rather than enhancing user acceptance. More recently, Blut et al.'s (2021) meta-analysis confirmed that the effects of anthropomorphism vary substantially across different types of agents and contexts, yet the sources of this heterogeneity remain insufficiently understood. These inconsistencies span the direction of effects (positive vs. negative), the boundary conditions under which effects emerge (visual vs. auditory cues; utilitarian vs. hedonic contexts), and the psychological mechanisms involved (social presence, warmth, eeriness).

We argue that a key reason for these inconsistencies is that prior research has predominantly treated anthropomorphism as a unidimensional construct, conflating qualitatively different types of human-like cues into a single manipulation or measure (Złotowski et al. 2014). When physical appearance cues, emotional expressions, and autonomous reasoning capabilities are bundled together or examined in isolation without reference to other dimensions, it becomes difficult to determine which specific facet of anthropomorphism drives observed effects, and under what conditions these effects may diverge or even conflict.

Several attempts have been made to organize the various anthropomorphic cues examined in prior research. Seeger et al. (2021) proposed a taxonomy that categorizes anthropomorphic cues of conversational agents into human-identity cues (e.g., avatars, names), verbal cues (e.g., emotional expressions, social dialog), and nonverbal cues (e.g., emoticons, turn-taking). While this taxonomy provides a useful inventory for chatbot designers, it is organized around the modality of the cue rather than the psychological mechanism it activates in users. As a result, cues that elicit the same type of human-like perception, such as emotional expressions (verbal) and emoticons (nonverbal), are placed in separate categories, making it difficult to predict their effects on user outcomes.

In the broader consumer behavior and branding literature, anthropomorphism has also been examined as a multi-dimensional phenomenon. Notably, Huang et al. (2020) developed a two-dimensional scale of brand anthropomorphism comprising “think” (cognitive capacity) and “feel” (emotional capacity), which bears conceptual similarity to the autonomous and emotional dimensions examined in the present research. Reavey et al. (2018) distinguished between subtle and overt humanization in advertising contexts, while Kim and McGill (2018) differentiated between designed and perceived anthropomorphism, highlighting that the same physical cue can elicit different psychological attributions depending on context. In the chatbot domain specifically, Chen et al. (2024) examined how visual appearance and conversational style independently influence user behavior, Alalwan et al. (2025) linked overall anthropomorphism to flow experience in online services, and Pozharliev et al. (2023) documented asymmetric blame attributions toward autonomous versus human agents. Reviewing this rapidly growing body of work, Greilich et al. (2025) explicitly called for a more systematic integration of anthropomorphism dimensions in future chatbot research.

Despite these advances, a comprehensive framework that (a) captures the full range of anthropomorphism dimensions relevant to service chatbots, including not only cognitive and emotional but also physical facets, (b) is grounded in the psychological process of attribution rather than the design modality of cues, and (c) specifies the distinct mechanisms through which each dimension affects user outcomes, remains lacking. This gap is consequential because, as we demonstrate in the present research, different anthropomorphism dimensions can not only produce different effects but can also interact in counterintuitive ways.

In this research, we conceptualize anthropomorphism as a multidimensional higher-order perception rather than a single construct. Specifically, we distinguish three forms of anthropomorphic attribution through which users perceive chatbots as humanlike: physical anthropomorphism (the perception that a chatbot possesses human-like physical or identity cues, such as a human name, avatar, or voice), emotional anthropomorphism (the perception that a chatbot can experience and respond to emotions), and autonomous anthropomorphism (the perception that a chatbot can understand and respond to user needs independently, flexibly, and adaptively, much like a human agent would). We develop the conceptual foundations of these dimensions in the theoretical background. We focus specifically on service chatbots that handle utilitarian requests, because the utilitarian nature of these interactions provides a theoretically meaningful context for examining the differential roles of warmth and competence as predicted by the stereotype content model (SCM; Liu et al. 2022). We examine how these dimensions influence customer satisfaction, which serves as the focal outcome variable across our studies.

Our framework extends prior multidimensional approaches, such as Huang et al.'s (2020) “think” and “feel” dimensions, in three important ways. First, it incorporates the physical dimension, which remains highly relevant for chatbots that employ avatars, human names, or embodied representations. Second, it is situated within the service chatbot context, where users'

evaluations are shaped by functional goals rather than brand relationships. Third, and most critically, it links each dimension to distinct evaluative mechanisms, perceived warmth and perceived competence, drawing on the SCM (Cuddy et al. 2008). Specifically, we argue that emotional and physical cues primarily evoke perceptions of warmth, whereas autonomous cues primarily signal competence. These warmth and competence perceptions, as the primary social evaluation dimensions, translate anthropomorphic perceptions into service evaluations. This integration allows us not only to predict which dimension drives satisfaction but also to explain why different dimensions produce different effects and how they may jointly shape user evaluations. Figure 1 presents the conceptual framework tested in this research.

We examine this framework through four studies using a mixed-method approach. Study 1 adopts an exploratory text-analytic approach to identify the key dimensions of chatbot anthropomorphism from consumers' own descriptions, using LLM-based topic modeling supplemented by human-coded validation; an exploratory method is necessary at this stage because the dimensionality of chatbot anthropomorphism has not yet been empirically established from the user's perspective. Study 2 experimentally tests the independent effects of these dimensions on customer satisfaction through the mediating mechanisms of perceived warmth and perceived competence, using a single-factor between-subjects design that allows us to isolate the unique contribution of each dimension. Study 3 examines the interaction between emotional and autonomous anthropomorphism using a 2×2 between-subjects design, which is required to test whether and how these dimensions jointly shape satisfaction in utilitarian service contexts. Study 4 replicates the 2×2 between-subjects design of Study 3, and explores how the interaction between emotional and autonomous anthropomorphism is translated into satisfaction through perceived warmth and competence.

This research makes three primary contributions to the literature. First, it moves beyond the prevailing unidimensional treatment of chatbot anthropomorphism by proposing and validating a three-dimensional framework grounded in users' psychological attributions rather than design-level cue taxonomies. In doing so, it provides a structured lens through which

previously mixed findings can be better understood: different studies have produced conflicting results not because anthropomorphism is inherently inconsistent in its effects, but because qualitatively distinct dimensions were confounded or examined in isolation. Second, by integrating the SCM into the study of chatbot anthropomorphism, this research specifies the evaluative mechanisms through which each dimension operates. We demonstrate that autonomous anthropomorphism uniquely enhances satisfaction through perceived competence, whereas emotional anthropomorphism enhances perceived warmth, which, in utilitarian service contexts, does not translate into higher satisfaction. This distinction offers both theoretical clarity and actionable guidance for chatbot design. Third, this research is among the first to examine the interactive effects of different anthropomorphism dimensions. Our finding that emotional anthropomorphism can attenuate the efficacy of autonomous capabilities challenges the common assumption that adding more human-like features is always beneficial and highlights the importance of considering dimensional interplay in both theory and practice.

2 | Theoretical Background

2.1 | The Theory of Anthropomorphism

Anthropomorphism refers to the process of attributing humanlike characteristics, both physical features (e.g., a face, eyes) and mental capacities (e.g., intentions, emotions, conscious awareness), to nonhuman entities (Waytz et al. 2010). The three-factor theory of anthropomorphism (Epley et al. 2007) identifies three determinants of this process: the accessibility of anthropocentric knowledge triggered by an agent's observable features and behaviors (elicited agent knowledge), the motivation to understand and predict the agent's actions (effectance motivation), and the desire for social connection (sociality motivation; Epley et al. 2008). While all three factors contribute to anthropomorphism, the elicited agent knowledge is most directly relevant to chatbot design, as it depends on the specific cues that the agent exhibits. The more an agent displays humanlike characteristics, the more likely users are to anthropomorphize it (Epley et al. 2007). In this research, we focus on these agent-side anthropomorphic cues and examine how different types of cues (physical, emotional, and autonomous)

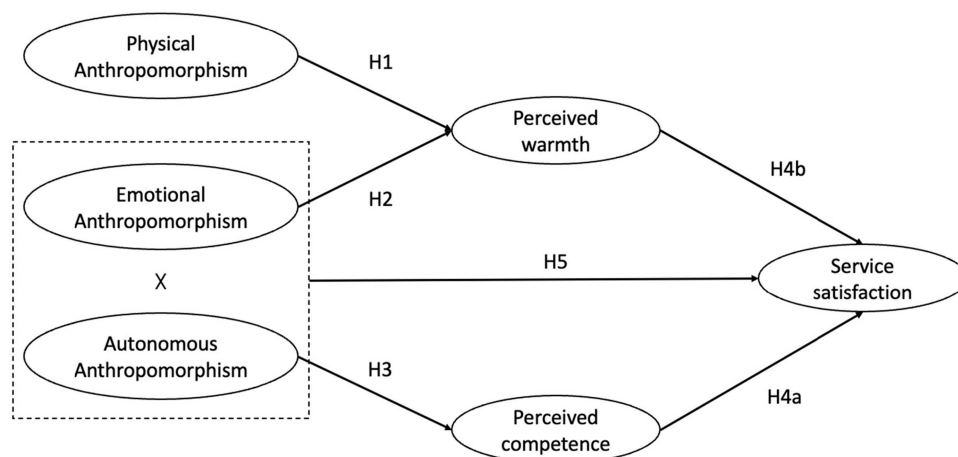


FIGURE 1 | Conceptual model.

elicit distinct anthropomorphic perceptions and, consequently, different user evaluations. These distinct cues correspond to different forms of anthropomorphic attribution that users make about the chatbot's human-likeness; we formally define these three dimensions after reviewing the relevant literature.

2.2 | Anthropomorphism in Human-Chatbot Interaction

Research on anthropomorphism in human-robot and human-chatbot interaction has generally followed two streams, mirroring the dichotomy between physical features and mental capacities inherent in the construct (Waytz et al. 2010). The first stream examines how humanlike physical features, such as facial appearance, body shape, gestures, and voice, shape user responses. A prominent theoretical account in this stream is the uncanny valley hypothesis (Mori et al. 2012), which proposes that increasingly humanlike appearance elicits positive responses up to a threshold, beyond which it triggers feelings of eeriness and discomfort. Empirical studies have confirmed that physical anthropomorphism can enhance perceived warmth and likability (Nunamaker et al. 2011; Blut et al. 2021), but may also increase perceived eeriness and threat when the humanlike appearance becomes too realistic (Kim et al. 2019; Mende et al. 2019). While many service chatbots are text-based interfaces without a full physical embodiment, physical anthropomorphism remains relevant in these contexts because many such chatbots employ identity-related cues such as human names, profile pictures, or avatars that convey a sense of human-like presence (Seeger et al. 2021).

The second stream focuses on the mental capacities attributed to artificial agents, which can be further divided into emotional and autonomous (or cognitive) capacities. Emotional anthropomorphism, the perception that an agent can express and respond to emotions, has been examined in healthcare (De Gennaro et al. 2020; Liu and Sundar 2018), education (Ahmad et al. 2019 February), and customer service contexts (Cheshin et al. 2018). Studies have generally found that emotional expressions enhance perceived intimacy, sincerity, and trust (Jiang et al. 2011; Ma and Wang 2021; Chandra et al. 2022). However, the effects are not uniformly positive: Stein and Ohler (2017) found that autonomously expressed empathy increased eeriness, suggesting that emotional cues may backfire under certain conditions.

Autonomous anthropomorphism, the perception that an agent possesses humanlike cognitive agency, including the ability to understand, reason, and respond to user needs independently and adaptively, has received increasing attention as AI systems become more capable. Beer et al. (2014) conceptualized robot autonomy as the capacity to sense the environment, plan, and act toward goals without external control. In service contexts, autonomous capabilities such as conversational relevance, adaptive responses, and proactive problem-solving have been found to enhance user engagement and perceived intelligence (Schuetzler et al. 2018; Clegg et al. 2020). At the same time, high levels of perceived autonomy can threaten users' sense of control and identity (Złotowski et al. 2017; Jörling et al. 2019), particularly in contexts involving embodied robots with physical agency.

Table 1 provides a summary of prior research on anthropomorphism across service and consumer contexts, including the dimensions examined, theoretical approaches, and key outcomes. As this overview illustrates, existing studies vary considerably in their conceptualization and theoretical grounding of anthropomorphism, underscoring the need for a more integrative framework.

Importantly, while these streams of chatbot anthropomorphism have generated valuable insights, they have largely proceeded in parallel. Studies typically manipulate one type of anthropomorphic cue (a humanlike avatar, an emotional expression, or an adaptive response style) without systematically accounting for the other dimensions. As we argued in the introduction, this fragmented approach is a key source of the mixed findings in the literature. In the present research, we address this limitation by examining all three dimensions, physical, emotional, and autonomous, both independently and in interaction.

2.3 | Defining the Three Dimensions of Chatbot Anthropomorphism

Building on the research reviewed above, we conceptualize chatbot anthropomorphism as a higher-order perception comprising three distinct but related dimensions. What unites these dimensions is the underlying psychological process, in that users in each case attribute a humanlike quality to a nonhuman agent (Epley et al. 2007). What distinguishes them is the target of that attribution, that is, which aspect of humanness the user perceives in the chatbot. This distinction follows the dichotomy between physical features and mental capacities noted above (Waytz et al. 2010), with mental capacities further differentiated into experiential and agentic facets, consistent with the two-dimensional structure of mind perception identified in prior research (Gray et al. 2007; Huang et al. 2020).

Specifically, physical anthropomorphism concerns attributions about the chatbot's outward humanlikeness, whereby users perceive that the agent possesses humanlike appearance or identity cues, such as a human name, avatar, voice, or visual representation. The target of the attribution is the agent's form and identity rather than its inner capacities. Emotional anthropomorphism concerns attributions about the chatbot's experiential capacity, whereby users perceive that the agent can feel, express, and respond to emotions. The target here is the agent's capacity for experience, that is, its perceived ability to have and share affective states. Autonomous anthropomorphism concerns attributions about the chatbot's agentic capacity, whereby users perceive that the agent can understand situations, reason, and act on user needs independently, flexibly, and adaptively. The target is the agent's capacity for agency, that is, its perceived ability to think and act on its own.

Two features of this conceptualization warrant emphasis. First, because it is organized around the psychological attribution made by the user rather than the design modality of the cue, the framework cuts across cue-based taxonomies (e.g., Seeger et al. 2021). A verbal cue such as empathetic language and a nonverbal cue such as an emoticon both feed the same emotional attribution, whereas two verbal cues, such as empathetic

TABLE 1 | Summary of prior research on anthropomorphism in service and consumer contexts.

References	Definition/ conceptualization	Dimension(s) examined	Theoretical approach	Moderators	Key Outcomes	Context
Araujo (2018)	Communicative agency framing	Human-like language cues (unidimensional)	CASA paradigm	—	Social presence (+); Attitudes (ns)	Online chat
Yuan and Dennis (2019)	Human-like appearance and behavior	Visual; Auditory anthropomorphism	Social response theory	Cue modality (visual vs. auditory)	Willingness to pay (+/–)	E-commerce
Stein and Ohler (2017)	Mind attribution to virtual characters	Autonomous empathy	Uncanny valley; Mind perception	Autonomy of emotional expression	Eeriness (+); Acceptance (–)	VR chat
Blut et al. (2021)	Human-like features in service agents	Physical anthropomorphism (meta-analytic)	Meta-analytic framework	Agent type; Service context	Social presence; Intelligence; Likability (+)	Multiple (robots, chatbots, AI)
Seeger et al. (2021)	Design-based cue taxonomy	Human-identity; Verbal; Nonverbal cues	CASA paradigm	—	Social presence; Trust	Text-based chatbot
Huang et al. (2020)	Brand anthropomorphism (mind attribution to brands)	Think (cognitive); Feel (emotional)	Mind perception theory	—	Moral judgment of brands	Brand perception
Kim and McGill (2018)	Product anthropomorphism	Designed vs. Perceived anthropomorphism	Social cognition	Financial status	Product evaluation	Consumer products
Reavey et al. (2018)	Advertising anthropomorphism	Subtle vs. Overt humanization	Persuasion theory	Cognitive busyness; Assertive language	Ad effectiveness	Advertising
Chen et al. (2024)	Anthropomorphic design cues in chatbots	Visual appearance; Conversational style	CASA paradigm	Dimension interaction (visual x conversational)	Information disclosure tendency; Perceived security	Text-based chatbot
Greilich et al. (2025)	Text-based AI chatbot anthropomorphism	Multiple (systematic review)	Multiple	Multiple	Consumer response (systematic review)	Text-based chatbot
Alalwan et al. (2025)	AI chatbot anthropomorphism and service quality	Overall anthropomorphism (unidimensional)	Flow theory	Telepresence	Flow experience; Emotional brand attachment; Brand loyalty	Online service
Pozharliev et al. (2023)	Agent autonomy in service outcomes	Autonomous vs. Human agent	Attribution theory	Service outcome (success vs. failure)	Blame attribution; Perceived competence; Satisfaction	Autonomous vehicles
Zlotowski et al. (2014)	Robot human likeness dimensions	Emotionality; Intelligence	Mind perception	—	Perceived human likeness	Robot (quiz context)
Kim et al. (2019)	Robot physical anthropomorphism	Physical appearance (human likeness)	Uncanny valley; SCM	—	Warmth (+); Liking (–); Eeriness (+)	Service robot
Cheshin et al. (2018)	Emotional expression intensity	Emotion intensity in service	Emotion as social information (EASI)	Perceived appropriateness	Trust; Satisfaction	Customer service

(Continues)

TABLE 1 | (Continued)

References	Definition/ conceptualization	Dimension(s) examined	Theoretical approach	Moderators	Key Outcomes	Context
Belanche et al. (2021)	Robot physical appearance in frontline	Physical appearance; Warmth; Competence	SCM; Humanness- Value-Loyalty	—	Warmth; Competence; Loyalty	Frontline service robot
Liu et al. (2022)	Robot appearance in service contexts	Friendly vs. Competent appearance	SCM	Service context (utilitarian vs. hedonic)	Usage intention	Tourism/service robot
Present study	Multidimensional anthropomorphic attribution	Physical; Emotional; Autonomous	SCM + EVT	Dimension interaction (Emo x Auto)	Warmth; Competence; Satisfaction	Service chatbot (utilitarian)

Abbreviations: CASA = Computers Are Social Actors, EASI = Emotions As Social Information, EVT = Expectancy Violation Theory, ns = not significant, SCM = Stereotype Content Model, (+) = positive effect, (–) = negative effect.
Source: Authors own work.

language and a proactive suggestion, feed different attributions. This attribution-based organization is what allows the framework to generate predictions about user outcomes, because it is the perception, not the cue itself, that activates downstream evaluative processes. Second, the three dimensions are conceptually distinct but not orthogonal. As facets of a common higher-order construct, they are expected to correlate positively, such that an agent perceived as humanlike in one respect is somewhat more likely to be perceived as humanlike in others (Epley et al. 2007). Indeed, in naturalistic descriptions of chatbot human-likeness (as we show in Study 1 below), users sometimes invoke multiple dimensions jointly, for example by expecting emotions to be conveyed through physical features such as voice. The claim of dimensionality, therefore, is not that the dimensions are unrelated, but that they are separable. They can be perceived, manipulated, and evaluated independently, and, as we demonstrate, they exert distinct effects on user outcomes through distinct evaluative mechanisms.

Our framework can be situated relative to existing multi-dimensional approaches, most notably the two-dimensional think-feel conceptualization of brand anthropomorphism (Huang et al. 2020). The emotional and autonomous dimensions in our framework correspond to the experience and agency facets of mind perception (Gray et al. 2007), which the think-feel approach also captures in the brand context. The physical dimension, however, is reducible to neither facet, and its inclusion is justified on three grounds. Conceptually, physical anthropomorphism concerns the agent's perceived form and identity rather than its mind; a chatbot can display a human name and avatar without being credited with any capacity to think or feel, and conversely. Theoretically, observable humanlike features constitute the most immediate trigger of elicited agent knowledge in the three-factor theory (Epley et al. 2007), so a framework that omitted them would exclude the very cues through which anthropomorphic attribution is most readily initiated. The physical dimension also connects our framework to phenomena that mind-based models cannot accommodate, such as uncanny valley effects driven by appearance (Mori et al. 2012; Kim et al. 2019). Empirically, the physical dimension emerged spontaneously as a distinct category in consumers' own descriptions of chatbot human-likeness in Study 1, and it reflects the most widespread anthropomorphic design practice in the chatbot industry, the use of human names, avatars, and voices. A two-dimensional mind-based framework would therefore be both conceptually incomplete and practically blind to a major class of design decisions.

2.4 | Warmth and Competence as Evaluative Mechanisms

To explain how different anthropomorphism dimensions translate into user evaluations, we draw on the stereotype content model (SCM; Fiske et al. 2002; Cuddy et al. 2008). The SCM posits that social perception is often organized along two fundamental dimensions: warmth and competence. Warmth reflects judgments about an agent's intentions and social orientation, that is, whether it is perceived as friendly, caring, sincere, and trustworthy. Competence reflects judgments about

an agent's ability, that is, whether it is perceived as capable, efficient, intelligent, and knowledgeable (Fiske et al. 2002). These two dimensions have been shown to account for a large proportion of variance in social judgments across diverse contexts, including interpersonal perception, group stereotypes, and brand evaluation (Aaker et al. 2010; Kervyn et al. 2012).

In the context of human-robot and human-chatbot interaction, the SCM has been increasingly applied to explain user responses. Reeves et al. (2020) demonstrated that users form warmth and competence impressions of social robots in a manner analogous to interpersonal perception. Belanche et al. (2021) found that the physical appearance of service robots influenced both warmth and competence perceptions, which in turn affected customer loyalty. Liu et al. (2022) showed that in utilitarian service contexts, competence perceptions were more predictive of favorable outcomes than warmth perceptions, whereas the reverse pattern held in hedonic contexts.

A critical insight from the SCM literature is that warmth and competence can operate in a compensatory relationship under certain evaluative conditions: increasing perceptions of one dimension may decrease perceptions of the other (Kervyn et al. 2010). This compensation effect has been documented in brand perception (Aaker et al. 2010), nonprofit versus for-profit evaluations (Kervyn et al. 2012), and robot perception (Liu et al. 2021). For service chatbots, this compensatory dynamic is particularly relevant: anthropomorphic cues that enhance perceived warmth (e.g., emotional expressions, friendly language) may inadvertently reduce perceived competence, and vice versa. This dynamic provides the theoretical basis for the interaction hypothesis we develop below.

Building on this framework, we propose that different anthropomorphism dimensions map onto different evaluative pathways. Physical and emotional anthropomorphism (cues related to human appearance, identity, and emotional expression) should primarily activate warmth perceptions, as they signal the chatbot's social presence and relational qualities rather than its functional capability. Autonomous anthropomorphism (cues related to adaptive reasoning, proactive problem-solving, and contextual understanding) should primarily activate competence perceptions, as they signal the chatbot's functional capability. In utilitarian service contexts, where customers interact with chatbots to accomplish specific tasks, competence perceptions should be more consequential for satisfaction than warmth perceptions (Liu et al. 2022).

2.5 | Expectancy Violations in Human-Chatbot Interaction

While the stereotype content model explains the evaluative mechanisms linking anthropomorphism dimensions to warmth and competence perceptions, expectancy violation theory (EVT; Burgoon 2016) helps explain why certain combinations of anthropomorphic cues may produce unexpected interaction effects. EVT posits that individuals hold baseline expectations about how interaction partners, including non-human agents, will behave. When these expectations are violated, the direction of the violation (positive or negative) shapes subsequent

evaluations. Positive violations, in which behavior exceeds expectations, tend to enhance evaluations; negative violations, in which behavior falls short of expectations, tend to diminish them.

In the context of service chatbots, users are likely to hold differentiated expectations for different types of human-like behavior. Given that most users understand they are interacting with an artificial agent, autonomous capabilities such as adaptive problem-solving and contextual understanding may exceed baseline expectations, constituting a positive violation that enhances satisfaction. In contrast, emotional expressions from a nonhuman agent may be perceived as inauthentic or incongruent with users' mental model of what a chatbot should do, particularly in task-oriented contexts where emotional displays are neither expected nor desired (Stein and Ohler 2017). When emotional and autonomous cues co-occur, the positive violation generated by autonomous capabilities may be attenuated by the ambiguity or negative valence of the emotional violation, reducing the overall benefit to satisfaction. This expectancy-based reasoning complements the compensatory logic of the SCM. While the SCM explains why warmth and competence perceptions may trade off at the evaluative level, EVT explains why certain combinations of anthropomorphic cues produce suboptimal outcomes at the experiential level: users' expectations are simultaneously confirmed and violated in ways that create evaluative tension rather than enhancement.

3 | Study 1: The Dimensions of Chatbot Anthropomorphism

3.1 | Data Collection and Procedure

To explore perceptions of chatbot anthropomorphism, we conducted an online survey using a UK panel called Prolific. First, the participants were asked to describe their experience of communicating with a conversational robot (chatbot), to ensure that they fully understood the context. To include all types of chatbots (text-, voice-, and video-based), we described each type before the survey. In the survey, we asked participants to elaborate as much as possible about "What makes such a chatbot like a human?" In total, 402 valid responses were collected. The respondents' mean age was 31.79 (SD = 10.80; minimum = 18; maximum = 69). There were more female participants (263 responses, 65.42%) than male participants (139 responses, 34.58%). Most participants had a bachelor's degree (146 responses, 36.32%), and a similar proportion of participants were high school graduates (132 responses, 32.84%); 72 participants (17.91%) reported having a college degree, and 49 participants (12.19%) reported holding a master's degree or above. The remaining three respondents (0.75%) received less than a high school education.

4 | Data Analysis

Topic modeling using two Large Language Models (LLM, i.e., ChatGPT 4 and Claude) was conducted to extract the significant chatbot dimensions of anthropomorphism from the answers to the open-ended question ("What makes a chatbot be like a human?") in the survey. Recent research has demonstrated that

LLM-based can match or exceed human performance in text annotation and classification tasks (Gilardi et al. 2023), and that LLM-derived thematic analyses show substantial agreement with human-coded interpretations (Tai et al. 2024; Castellanos et al. 2025). Using multiple LLM-based in parallel allows for cross-validation of the extracted themes, reducing the risk that the identified dimensions are artifacts of any single model's processing tendencies.

First, the qualitative data obtained from the open-ended responses was collated and prepared for analysis. The text was consolidated into a single unit and the input was verified to comply with the token limit of the LLM-based. Then, a prompt was devised to instruct the LLM-based to identify the principal topics present in the given text (see Appendix SI for the prompts used). The prompt included detailed instructions for the LLM-based to generate a structured output comprising the title and definition for each identified topic. The selected LLM-based (GPT 4 and Claude) were then prompted separately with the prepared text and the instructions to identify the main topics. The LLM-based produced their respective outputs, enumerating the identified topics with their titles and definitions. After that, a comparative analysis was conducted to assess the consistency of topic extraction across the various LLM-based. In particular, the number of identical topics identified by GPT 4 in comparison to those identified by Claude was calculated. The discrepancies in topic granularity were addressed by means of the comparative analysis. The categorizations were refined through the merging of related topics and the splitting of broad topics into more specific ones, thereby ensuring coherence and alignment with the qualitative data. A comparison of the topic modeling results from the two LLM-based revealed a notable consistency in the identified topics and their descriptions. Although there were some discrepancies in the level of detail and the terminology used to describe the topics, the overarching themes and issues highlighted by the participants were largely consistent between the two models.

To further validate the LLM-based topic modeling results, we randomly selected 100 responses from the sample and asked

two independent coders to classify each response according to whether it mentioned physical, emotional, and/or autonomous anthropomorphism, based on the definitions of these dimensions. Inter-rater reliability between the two human coders and the LLM output was assessed using Krippendorff's alpha. The results indicated strong agreement across all three dimensions: $\alpha = 0.828$ for physical anthropomorphism, $\alpha = 0.906$ for emotional anthropomorphism, and $\alpha = 0.821$ for autonomous anthropomorphism. These results indicate that the LLM-derived dimensions are consistent with human coders' understanding, supporting the validity of the topic modeling approach.

5 | Results

Claude identified three main topics related to chatbot anthropomorphism: *Autonomous anthropomorphism*, *Emotional anthropomorphism*, and *Physical anthropomorphism*, each with multiple subtopics (See Tables 2 and 3). *Thinking skills*, *understanding language*, *accurate response*, and *task performance* are the subtopics of autonomous anthropomorphism. *Thinking skills* encompass essential cognitive abilities, including problem-solving, decision-making, adaptability, and learning. *Understanding language* focuses on the chatbot's proficiency in understanding speech, questions, instructions, and context. *An accurate response* is about the chatbot's ability to provide appropriate and correct answers. It is closely related to the other subtopics within autonomous anthropomorphism, as accurate responses are a result of good thinking skills and language understanding. *Task performance* pertains to the chatbot's ability to carry out human-like jobs and tasks, further highlighting its autonomous capabilities.

As for emotional anthropomorphism, subtopics include *communication style*, *emotion*, and *personality*. *Communication style* covers the use of humor and the display of empathy in communication. *Emotion and personality* involve the chatbot's display of emotions, having a distinct personality, and showing care, all of which enhance the emotional connection between the user and the chatbot.

TABLE 2 | Topic modeling results from Claude.

Topic	Subtopic	Definition
Autonomous anthropomorphism	Thinking skills	This topic covers abilities like problem-solving, decision making, adaptability, and learning.
	Understanding language	This covers abilities like understanding speech, questions, instructions, context.
	Accurate response	This new topic covers responding appropriately, having accurate responses.
	Task performance	This topic is about performing human jobs and tasks.
Emotional anthropomorphism	Communication style	This topic covers abilities like using humor, and displaying empathy.
	Emotion and personality	This topic involves displaying emotions, having a personality, and showing care.
Physical anthropomorphism	Human-like appearance	This topic includes having human-like features, movements, gestures, and expressions.

Source: Authors own work.

TABLE 3 | Topic modeling results from GPT 4.0.

Topic	Subtopic	Definition
Autonomous anthropomorphism	Decision-making and Autonomy	The chatbot's ability to make decisions independently, prioritize tasks, and solve problems without human intervention. This includes understanding contradictory commands and choosing the most appropriate action.
	Communication Skills	The ability of the robot to engage in meaningful conversations with humans. This includes understanding context, asking clarifying questions, and providing relevant and coherent responses.
	Responsiveness and Adaptability	The speed and appropriateness of the robot's reactions to stimuli, including its ability to adapt to new situations or unexpected challenges.
	Task Performance	The capability of the robot to perform tasks that are traditionally done by humans, either physical tasks like cleaning or intellectual tasks like problem-solving.
Emotional anthropomorphism	Emotional Intelligence	The ability to understand, interpret, and respond to human emotions. This includes showing empathy, recognizing emotional cues, and adapting behavior based on the emotional context.
	Social Skills	The ability to engage in social interactions, understand social cues, and behave in a manner that is considered socially acceptable or relatable.
	Humor and Personality	The extent to which the chatbot can display a sense of humor or a distinct personality, making interactions more enjoyable and less formulaic.
Physical anthropomorphism	Physical Appearance	The extent to which the chatbot mimics human physical features, such as having a face, eyes, or limbs. This also includes the robot's ability to make facial expressions or other human-like gestures.
	Human movement	The fluidity and naturalness of the chatbot's physical movements, including smiling, gesturing, and other forms of locomotion.
	Voice and Tone	The quality of the chatbot's voice, including its ability to mimic human intonation, accent, and emotional tone, making interactions more relatable and less mechanical.

Source: Authors own work.

Physical anthropomorphism only has one subtopic, which is human-like appearance. It includes having human-like features, movements, gestures, and expressions, suggesting that the physical appearance of the chatbot is a crucial factor in its anthropomorphism.

GPT 4.0 also identified the same three main topics but with different subtopics. Subtopics of autonomous anthropomorphism identified by GPT 4.0 are *decision-making and autonomy*, *communication skills*, *responsiveness and adaptability*, as well as *task performance*. *Decision-making and autonomy* focus on the chatbot's independent decision-making ability, task prioritization, and problem-solving without human intervention. It also includes handling contradictory commands, which is a complex aspect of autonomous behavior. *Communication skills* emphasize the robot's ability to have meaningful conversations. This involves understanding context, asking clarifying questions, and providing relevant and coherent responses. *Responsiveness and*

adaptability refer to the speed and appropriateness of the chatbot's reactions to various stimuli, as well as its ability to adapt to new or unexpected situations. *Task performance* is related to the chatbot's capacity to perform tasks typically done by humans, both physical and intellectual.

GPT 4.0 identified 3 subtopics for emotional anthropomorphism, which are *emotional intelligence*, *social skills*, and *humor and personality*. *Emotional intelligence* is about the chatbot's ability to understand, interpret, and respond to human emotions. It includes showing empathy, recognizing emotional cues, and adapting behavior based on the emotional context, which is crucial for building a connection with users. *Social skills* pertain to the chatbot's ability to engage in social interactions, understand social cues, and behave in a socially acceptable manner. *Humor and personality* focus on the chatbot's ability to display a sense of humor and a distinct personality, making interactions more enjoyable.

More nuanced subtopics of physical anthropomorphism were identified by GPT 4.0 than Claude. *Physical appearance* is about the chatbot's mimicry of human physical features, such as having a face, eyes, or limbs, and its ability to make facial expressions or other human-like gestures. *Human movement* focuses on the fluidity and naturalness of the chatbot's physical movements, including smiling, gesturing, and locomotion. *Voice and tone* pertain to the quality of the chatbot's voice, including its ability to mimic human intonation, accent, and emotional tone.

5.1 | Discussion

Study 1 explored people's perception of "what makes a chatbot be like a human." Topic modeling of the participants' answers to this question with Claude and GPT 4.0 both identified the same overarching themes of *Autonomous anthropomorphism*, *Emotional anthropomorphism*, and *Physical anthropomorphism*. This indicates that these are fundamental aspects of chatbot anthropomorphism that are consistently recognized by different language models. Based on previous studies and the findings of Study 1, we define physical anthropomorphism as the perception that a chatbot possesses human-like physical or identity cues, such as a human name, avatar, voice, or other human-representing features. Emotional anthropomorphism is defined as the perception that a chatbot can experience and respond to emotions. Autonomous anthropomorphism refers to the perception that a chatbot can understand and respond to user needs independently, flexibly, and adaptively much like a human agent would. These anthropomorphism dimensions induced from survey respondents' answers serve as a basis for further research in this area.

The disparity between Claude's and GPT 4.0's topic modeling results mainly relates to the focus and categorization of subtopics. Claude's subtopics within autonomous anthropomorphism are more general cognitive and language-related skills, while GPT 4.0's subtopics emphasize decision-making autonomy. For emotional anthropomorphism, Claude focused more on the display of emotions and personality, while GPT 4.0 emphasized emotional intelligence. In physical anthropomorphism, GPT 4.0 further divided the topic into appearance, movement, and voice, while Claude grouped these under a single subtopic of human-like appearance.

It is worth noting that these three dimensions overlap in the participants' answers. For example, participants mentioned that the emotions of a chatbot should be expressed naturally through its physical features (e.g., "Nuances in its voice that express emotions"). Also, the emotional capacity and autonomy of chatbots were frequently connected in the responses. One participant mentioned that a human-like chatbot should be able to "tell and understand jokes," and this requires the chatbot to have both considerable autonomy and emotional capacity. The overlap between different dimensions of chatbot anthropomorphism implies the need to investigate the interactive effects of these dimensions.

We note that Study 1 serves a qualitative, exploratory function. It establishes that the three dimensions emerge spontaneously

in consumers' own descriptions of chatbot human-likeness, and the human coding analysis validates the LLM-based topic modeling of this qualitative material rather than the dimensions' factor structure. The quantitative test of whether the three perceived dimensions form empirically separable factors is provided in the following experimental studies.

5.2 | Hypothesis Development

Based on the theoretical framework outlined above and the three dimensions identified in Study 1, we develop the following hypotheses regarding the effects of physical, emotional, and autonomous anthropomorphism on customer satisfaction with service chatbots. Consistent with the SCM, our predictions concern the primary evaluative pathway activated by each dimension. We do not predict that each dimension affects one perception exclusively. At the level of a single evaluated target, warmth and competence judgments are typically positively correlated, with compensatory relations emerging mainly when perceivers must reconcile mixed or competing signals (Kervyn et al. 2010). The distinctness of the dimensions, therefore, lies in which perception carries their effect forward to satisfaction, rather than in selective effects on one perception alone.

The effect of physical anthropomorphism. Physical anthropomorphism, conveyed through cues such as humanlike avatars, human names, and visual representations, signals social presence and human identity. According to the three-factor theory of anthropomorphism, such observable humanlike features are among the most immediate triggers of anthropocentric knowledge, leading users to draw on schemas of human interaction when evaluating the agent (Epley et al. 2007). In the SCM framework, these cues are primarily associated with warmth, as they convey the chatbot's social and relatable qualities rather than its functional capabilities. Prior research supports this link. Blut et al. (2021) found that physical anthropomorphism was positively related to social presence and likability, and Kim et al. (2019) demonstrated that humanlike physical features enhanced perceived warmth. Because such cues convey social presence rather than functional capability, their evaluative significance should lie primarily in warmth perceptions. Thus:

5.3 | H1: Physical Anthropomorphism of a Service Chatbot Positively Affects Perceived Warmth

The effect of emotional anthropomorphism. Emotional anthropomorphism, conveyed through cues such as empathetic language, emojis, and expressions of care, signals the chatbot's affective and relational capacities. Like physical cues, emotional cues align with the warmth dimension of the SCM, as they convey sincerity, friendliness, and interpersonal orientation (Cheshin et al. 2018; Ma and Wang 2021). Self-disclosure and emotional expression have been linked to increased perceived intimacy and trust in computer-mediated communication (Jiang et al. 2011; Chandra et al. 2022), and emotional support from digital agents has been shown to shape customer responses in technology-mediated services (Gelbrich et al. 2021). Emotional cues, by contrast, carry little direct

information about the chatbot's capacity to solve problems or perform tasks efficiently, so their evaluative significance should likewise lie primarily in warmth perceptions. Thus:

5.4 | H2: Emotional Anthropomorphism of a Service Chatbot Positively Affects Perceived Warmth

The effect of autonomous anthropomorphism. Autonomous anthropomorphism, conveyed through cues such as adaptive responses, proactive suggestions, and contextual understanding, signals the chatbot's cognitive and functional capabilities. Autonomy has been conceptualized as the capacity to sense, plan, and act toward goals without external control (Beer et al. 2014), and autonomous capabilities such as conversational relevance and proactive problem-solving have been found to enhance perceived intelligence and engagement (Schuetzler et al. 2018; Clegg et al. 2020). In the SCM framework, these cues map onto the competence dimension, as they convey intelligence, efficiency, and problem-solving ability. Thus:

5.5 | H3a: Autonomous Anthropomorphism of a Service Chatbot Positively Affects Perceived Competence

Importantly, autonomous behavior may also elevate warmth perceptions. An agent that proactively attends to a user's needs signals not only ability but also benevolent intent, which is the core of the warmth dimension (Fiske et al. 2002). This possibility is consistent with the conceptual status of the dimensions as positively correlated facets of a higher-order construct, and it implies that selective effects on a single perception are not the appropriate test of dimensional distinctness. The critical question for our framework is therefore not whether autonomy influences warmth, but which perception transmits its effect to satisfaction. The SCM literature suggests that the evaluative weight of warmth and competence depends on the perceiver's goals in the interaction (Cuddy et al. 2008). In utilitarian service contexts, where the chatbot's primary role is to address functional needs, competence is the diagnostic criterion against which the interaction is evaluated, whereas warmth is peripheral to the user's goal (Liu et al. 2022). Users in such contexts may register a chatbot's warmth, yet this perception carries little weight in their overall evaluation because it is not relevant to the task they came to accomplish. Accordingly, even if autonomous anthropomorphism enhances both competence and warmth perceptions, its satisfaction benefits should flow through the competence pathway, while any concurrent warmth gains should remain evaluatively inert. Thus:

H3b: Autonomous anthropomorphism of a service chatbot positively affects customer satisfaction through perceived competence rather than perceived warmth.

The link between warmth, competence, and satisfaction. The SCM posits that both warmth and competence perceptions influence overall evaluations, and their downstream consequences have been documented across interpersonal, brand, and human-machine contexts (Cuddy et al. 2008;

Aaker et al. 2010; Kervyn et al. 2012). In service contexts specifically, perceived competence has been consistently linked to satisfaction, as it reflects the agent's ability to fulfill the customer's functional needs (Belanche et al. 2021). Evidence from automated agents points in the same direction, with competence perceptions shaping satisfaction with autonomous service providers (Pozharliev et al. 2023). Perceived warmth can also contribute to satisfaction, particularly when relational or emotional needs are salient in the encounter (Gao and Mattila 2014), although its weight is expected to be lower in utilitarian settings (Liu et al. 2022). We therefore predict positive effects of both perceptions on satisfaction, while our framework implies that the competence link should be the stronger of the two in the present context. Thus:

H4a: Perceived competence of a service chatbot positively affects customer satisfaction.

H4b: Perceived warmth of a service chatbot positively affects customer satisfaction.

The interaction between emotional and autonomous anthropomorphism. Based on both the compensatory logic of the SCM (Kervyn et al. 2010) and expectancy violation theory (Burgoon 2016), we predict that emotional anthropomorphism will attenuate the positive effect of autonomous anthropomorphism on satisfaction. From the SCM perspective, the simultaneous display of warmth-signaling and competence-signaling cues confronts users with mixed signals that must be reconciled. Under such conditions, warmth-enhancing cues can dilute the diagnostic weight of competence cues (Kervyn et al. 2010). From the EVT perspective, in utilitarian service contexts, emotional expressions from a chatbot may constitute an ambiguous or negative expectancy violation that interferes with the positive violation generated by autonomous capabilities. Together, these arguments suggest that when a chatbot simultaneously displays high emotional and high autonomous anthropomorphism, the competence-based satisfaction benefits of autonomy may be attenuated.

We focus this interaction hypothesis on emotional and autonomous anthropomorphism rather than including physical anthropomorphism for two reasons. First, Study 2 provides initial evidence that physical anthropomorphism does not significantly affect satisfaction either directly or indirectly, suggesting limited potential for meaningful interaction effects. Second, the theoretical tension between warmth-signaling and competence-signaling cues is most acute between emotional and autonomous anthropomorphism, as these dimensions most directly activate the competing evaluative pathways described by the SCM. Thus:

H5: There is an interaction effect between emotional and autonomous anthropomorphism on customer satisfaction; specifically, the positive effect of autonomous anthropomorphism on satisfaction is attenuated when emotional anthropomorphism is high.

Study 2: The Main Effect of Autonomous, Emotional, and Physical Anthropomorphism.

6 | Research Method

Study 2 employed a single-factor experiment design (no vs. autonomous vs. emotional vs. physical anthropomorphism). In this experimental study, we compared the effect of different dimensions of anthropomorphism on consumer satisfaction.

We designed four versions of conversations between a user and a text-based chatbot in a flight booking scenario (see Appendix SII for the stimuli). In the experiment, participants were exposed to the screenshots of these conversations. All conversations followed the same flight booking procedure, and the chat interface layout remained consistent across conditions. The only variations were the specific anthropomorphic elements of the chatbot manipulated in each condition.

In the control condition (no anthropomorphism), the chatbot was identified simply as “TripPlus” and provided direct, functional responses without any anthropomorphic features. The communication style was standardized and template-like, focusing purely on the task requirements without additional engagement elements.

In the autonomous anthropomorphism condition, the chatbot demonstrated proactive behavior by independently searching for flight alternatives and providing suggestions without explicit user requests. When users mentioned date preferences, the chatbot actively explored multiple options and offered strategic suggestions about timing and pricing. The responses showed adaptive behavior by adjusting recommendations based on user inputs while maintaining a function-oriented but initiative-taking communication approach.

For the emotional anthropomorphism condition, the chatbot incorporated emotional elements through both text and emojis. The responses included emotional expressions with corresponding emojis to convey feelings such as happiness and concern. The conversational tone remained consistently warm and enthusiastic, using exclamation marks for emphasis and expressing genuine happiness in serving the customer. The emotional engagement was evident in phrases expressing care and understanding.

In the physical anthropomorphism condition, the chatbot was personified with a human-like nickname and a human avatar as the profile picture. The chatbot introduced itself as a personal assistant and maintained consistent use of first-person pronouns throughout the conversation. The human-like identity was reinforced through both visual and textual elements, creating a more personalized interaction experience.

We recruited 213 participants from an online panel, WJX. The average age was 31.7 (SD = 8.23); 44.6% of the participants were female ($n = 95$). In terms of education, 16.8% had not obtained a bachelor's degree, 76.5% had a bachelor's degree, and 6.6% had earned a master's degree or above.

Autonomous anthropomorphism was measured with three author-developed items informed by the conceptual definition of agent autonomy in Beer et al. (2014) (“This chatbot is capable of planning,” “This chatbot responds flexibly to my questions,”

“This chatbot has the ability to control its own actions”; $\alpha = 0.68$). Emotional anthropomorphism was measured with three author-developed items informed by Cheshin et al. (2018) (“This chatbot displays emotions,” “I can perceive this chatbot's emotions,” “The emotions this chatbot displays are intense”; $\alpha = 0.83$). Perceived physical anthropomorphism was measured with a single item assessing perceived physical human-likeness: “The physical appearance of the service agent looks like a human” (“This chatbot has the appearance of a real person”). Perceived warmth was measured with three items ($\alpha = 0.91$) and perceived competence with four items ($\alpha = 0.84$) (Fiske et al. 2002). Satisfaction was measured with three items ($\alpha = 0.81$). All items were on 7-point scales. Full wordings appear in Table SA1 in Appendix II.

6.1 | Data Analysis

Manipulation checks. The results showed that, in comparison with the chatbot in the control condition, participants perceived the autonomous chatbot as more autonomous ($M_{\text{control}} = 4.30$, $SD = 1.38$, $M_{\text{autonomous}} = 5.46$, $SD = 0.84$, $t(100) = -5.06$, $p < 0.001$); the emotional chatbot as more emotional ($M_{\text{control}} = 2.46$, $SD = 1.22$, $M_{\text{emotional}} = 3.43$, $SD = 1.23$, $t(105) = -4.07$, $p < 0.001$); and the physical human-like chatbot as more human-like ($M_{\text{control}} = 2.70$, $SD = 1.53$, $M_{\text{physical}} = 3.98$, $SD = 1.70$, $t(108) = -4.16$, $p < 0.001$). As such, all manipulations were successful.

Cross-dimensional manipulation checks. To assess dimensional separability, we examined whether each manipulation selectively influenced only its focal dimension. Table SA2 in Appendix II reports the means and standard deviations for all three manipulation check measures across all four conditions. One-way ANOVAs with Bonferroni post-hoc tests were conducted for each manipulation check measure.

For the physical manipulation check, the one-way ANOVA was significant ($F(3, 209) = 7.04$, $p < 0.001$). Bonferroni post-hoc comparisons showed that the physical condition scored significantly higher than the control condition ($p < 0.001$), the autonomous condition ($p = 0.020$), and the emotional condition ($p = 0.006$). Critically, neither the autonomous condition ($p = 1.000$) nor the emotional condition ($p = 1.000$) differed significantly from control on this measure, confirming that the physical manipulation selectively increased perceived physical anthropomorphism.

For the emotional manipulation check, the one-way ANOVA was significant ($F(3, 209) = 5.70$, $p < 0.001$). The emotional condition scored significantly higher than the control condition ($p < 0.001$). The physical condition did not differ significantly from control ($p = 0.113$). However, the autonomous condition also scored significantly higher than control on the emotional measure ($p = 0.027$), suggesting some cross-dimensional spillover. This partial overlap is consistent with the theoretical observation that a chatbot displaying proactive, adaptive behavior may also be perceived as somewhat more emotionally engaged. Importantly, the emotional condition's mean on this measure ($M = 3.43$) was numerically higher than the

autonomous condition's ($M = 3.17$), and the autonomous condition did not differ significantly from the emotional condition ($p = 1.000$).

For the autonomous manipulation check, the one-way ANOVA was significant ($F(3, 209) = 9.89, p < 0.001$). The autonomous condition scored significantly higher than control ($p < 0.001$) and significantly higher than the physical condition ($p = 0.008$). The physical condition did not differ significantly from control ($p = 0.157$). However, the emotional condition also scored significantly higher than control on the autonomous measure ($p = 0.037$), indicating some degree of cross-dimensional influence. This pattern may reflect that emotionally expressive chatbots are also perceived as somewhat more responsive and adaptive, consistent with the conceptual proximity of emotional engagement and cognitive responsiveness in chatbot interactions.

These results indicate that each manipulation produced its strongest effect on its focal dimension, while the physical manipulation showed clean discriminant validity with no significant cross-dimensional effects. The emotional and autonomous manipulations showed some mutual spillover, which is consistent with the theoretical position that anthropomorphism dimensions, while conceptually distinct, share a common higher-order construct (Epley et al. 2007). More generally, spillover at the level of manipulations should be distinguished from inseparability at the level of perceptions. Anthropomorphic cues co-occur in naturalistic chatbot design, so a manipulation of one dimension may plausibly move perceptions of another; whether the underlying perceptions themselves are empirically separable is a measurement question, which we address through the factor-analytic evidence reported below. Importantly, the mediation analyses reported below provide a complementary test of dimensional specificity by

examining whether each dimension operates through its theoretically predicted evaluative mechanism.

Satisfaction. The results of one-way ANOVA suggested that participants had different levels of satisfaction with the different conditions ($F(3, 209) = 5.77, p < 0.001, \eta^2 = 0.08$, see Figure 2). A post hoc analysis (using the Bonferroni correction) demonstrated that, compared to the control condition, participants in the autonomous anthropomorphism condition exhibited higher service satisfaction ($M_{\text{control}} = 4.97, SD = 1.08, M_{\text{autonomous}} = 5.80, SD = 0.97, p < 0.001$). However, we did not observe significant differences between the control condition and the emotional anthropomorphism condition ($M_{\text{emotional}} = 5.35, SD = 1.00, p = 0.34$) or between the control condition and the physical anthropomorphism condition ($M_{\text{physical}} = 5.29, SD = 0.95, p = 0.59$).

Effects on perceived warmth and competence. To examine the effects of the anthropomorphism manipulations on the proposed mediating variables, one-way ANOVAs were conducted with perceived competence and perceived warmth as dependent variables.

For perceived competence, the one-way ANOVA was significant ($F(3, 209) = 6.15, p < 0.001, \eta^2 = 0.08$). Bonferroni post-hoc comparisons showed that the autonomous condition ($M = 5.81, SD = 0.91$) scored significantly higher than the control condition ($M = 4.99, SD = 1.08, p < 0.001$) and the physical condition ($M = 5.14, SD = 0.99, p = 0.005$), supporting H3a. Neither the emotional condition ($M = 5.28, SD = 1.08, p = 0.879$) nor the physical condition ($M = 5.14, SD = 0.99, p = 1.000$) differed significantly from control. For perceived warmth, the one-way ANOVA was significant ($F(3, 209) = 8.95, p < 0.001, \eta^2 = 0.11$). The emotional condition ($M = 5.06, SD = 1.13$) scored significantly higher than the control condition ($M = 4.08, SD = 1.46$,

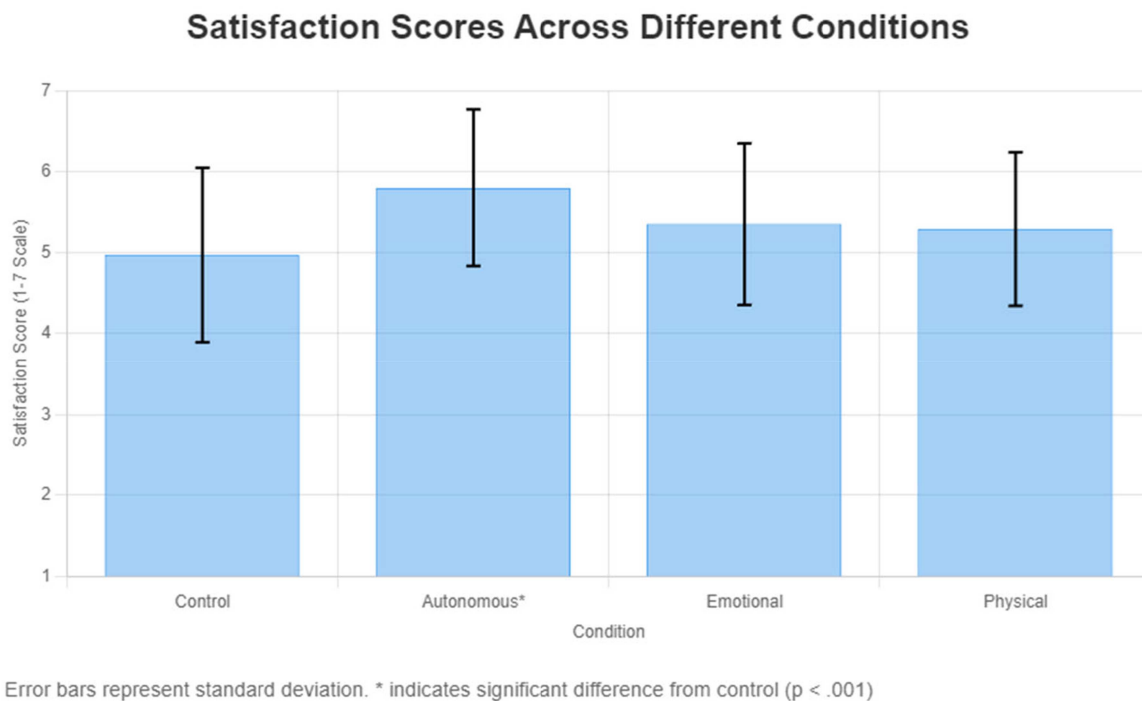


FIGURE 2 | The effect of different anthropomorphism dimensions on satisfaction, Study 2.

$p < 0.001$), supporting H2. The physical condition ($M = 4.65$, $SD = 1.35$) showed a directionally positive but nonsignificant difference from control ($p = 0.136$); H1 was therefore not supported. We attribute this to the relative weakness of name-and-avatar cues in a static, text-based interface. Notably, the autonomous condition ($M = 5.31$, $SD = 1.20$) also scored significantly higher than control on warmth ($p < 0.001$). This pattern is consistent with the position developed in our hypotheses that autonomous behavior signals benevolent intent alongside ability, and it underscores that the critical test of our framework lies not in selective effects on perceptions but in which perception transmits the effect to satisfaction (H3b), examined next.

Mediation analysis. To test the mediation hypotheses, we used PROCESS for R 0.2.6 (Model 4, bootstrap samples = 5000). As the anthropomorphism type was a multi-categorical variable, we used the indicator coding method and created three dummy variables comparing each anthropomorphism condition with the control condition. Compared to no anthropomorphism, autonomous anthropomorphism led to higher perceived competence, which in turn increased service satisfaction (point estimate = 0.55, $se = 0.15$, 95% CI: [0.27, 0.85]). The indirect effect of autonomous anthropomorphism on satisfaction through perceived warmth was not significant (point estimate = 0.07, $se = 0.06$, 95% CI: [−0.05, 0.20]). To test formally whether the two pathways differ, we computed the contrast between the indirect effect of autonomous anthropomorphism through perceived competence and through perceived warmth. The contrast was significant (difference = 0.48, $SE = 0.17$, 95% CI: [0.18, 0.83]). The indirect effect through competence was reliably larger than the indirect effect through warmth. Thus, although the autonomous manipulation elevated both competence and warmth perceptions, only the competence pathway carried the effect to satisfaction, supporting H3b. The indirect effect of emotional anthropomorphism on satisfaction was not significant through perceived warmth (point estimate = 0.06, $se = 0.05$, 95% CI: [−0.04, 0.18]) or through competence (point estimate = 0.19, $se = 0.15$, 95% CI: [−0.07, 0.49]). Physical anthropomorphism had no significant indirect effect through warmth (point estimate = 0.03, $se = 0.03$, 95% CI: [−0.02, 0.11]) or competence (point estimate = 0.10, $se = 0.14$, 95% CI: [−0.17, 0.36]).

Measurement model. Because our framework rests on the claim that the anthropomorphism dimensions are empirically separable from one another and from the evaluative mediators, we conducted a confirmatory factor analysis on the sixteen multi-item indicators, specifying five correlated factors: autonomous anthropomorphism (three items), emotional anthropomorphism (three items), perceived warmth (three items), perceived competence (four items), and satisfaction (three items). Physical anthropomorphism, measured with a single item, could not be modeled as a latent factor and was omitted. The five-factor model fit the data well ($\chi^2(94) = 130.66$, $CFI = 0.981$, $TLI = 0.976$, $RMSEA = 0.043$, $SRMR = 0.047$), and all loadings were significant ($\lambda = 0.45$ – 0.92). Critically, the five-factor model outperformed every theoretically motivated alternative: merging autonomous anthropomorphism with perceived competence ($\Delta\chi^2(4) = 23.99$, $p < 0.001$), perceived competence with satisfaction ($\Delta\chi^2(4) = 21.61$, $p < 0.001$), emotional

anthropomorphism with perceived warmth ($\Delta\chi^2(4) = 180.88$, $p < 0.001$), the two anthropomorphism dimensions with each other ($\Delta\chi^2(4) = 165.61$, $p < 0.001$), or all items onto a single factor ($\Delta\chi^2(10) = 509.93$, $p < 0.001$; see Table SA3 in Appendix II). At the same time, the latent correlations between autonomous anthropomorphism and perceived competence ($\phi = 0.89$) and between competence and satisfaction ($\phi = 0.90$) were high, and the average variance extracted for autonomous anthropomorphism fell below the conventional 0.50 threshold ($AVE = 0.43$), reflecting the fact that the autonomy items partly index competent task performance.

6.2 | Discussion

Study 2 examined the effects of the three anthropomorphism dimensions identified in Study 1 on customer satisfaction, as mediated by perceived warmth and perceived competence. Three findings stand out. First, only autonomous anthropomorphism significantly increased satisfaction relative to the control condition, and this effect operated through perceived competence, supporting H3a, H3b, and H4a. Second, emotional anthropomorphism enhanced perceived warmth, supporting H2, but this warmth did not translate into higher satisfaction; the indirect effect through warmth was not significant, consistent with the position that warmth carries limited evaluative weight in utilitarian service contexts (Liu et al. 2022). Third, H1 was not supported, as physical anthropomorphism produced no significant effect on warmth, satisfaction, or competence. Rather than treating these results as partial failures of the framework, we note that they sharpen its central claim. The dimensions did not differ merely in which perceptions they elevated; they differed in which perceptions carried evaluative consequences. This distinction between perceptual activation and evaluative transmission motivates the design of Study 3, which examines how emotional and autonomous anthropomorphism jointly shape satisfaction.

7 | Study 3: The Interaction Effect of Autonomous and Emotional Anthropomorphism

7.1 | Research Method

To test hypothesis 5, we conducted a 2 (autonomous anthropomorphism: high vs. low) $\times$ 2 (emotional anthropomorphism: high vs. low) between-subject experiment. In this study, respondents were told to imagine that they would like to purchase a pair of sneakers and that they were going to chat with a service chatbot to find out more information. Next, a screenshot of the conversation was shown (see Appendix III for the stimuli used in Study 3). In the high autonomous anthropomorphism condition, the chatbot exhibited behaviors that demonstrated a high level of independence and adaptability. For instance, the chatbot responded to user queries with contextually relevant follow-up questions, provided suggestions without direct prompts, and adapted its responses based on user input. In contrast, the low autonomous anthropomorphism condition featured a chatbot that gave more rigid, pre-scripted responses, demonstrating limited autonomy by sticking closely to a pre-defined set of responses without further engagement. For the high emotional anthropomorphism condition, the chatbot's

responses included emotive language and expressions, such as using emoji or words that conveyed empathy, humor, and enthusiasm. For example, in response to a user inquiry, the chatbot might say, “I totally understand!: blush: Let me help you with that!”. The low emotional anthropomorphism condition, on the other hand, involved responses that were strictly factual and devoid of emotional cues, such as simply stating, “Here is the information you requested.” These manipulations were intended to vary the levels of perceived emotional and autonomous anthropomorphism independently.

Study 3 was specifically designed to test the interaction hypothesis (H5) on satisfaction. As the theoretical rationale for this interaction is grounded in both the compensatory logic of the SCM and expectancy violation theory, the study focuses directly on satisfaction as the outcome rather than re-measuring the full mediating model tested in Study 2.

We recruited 343 respondents through an online panel, WJX. The mean age of the participants was 32.8 (SD = 10.4), and 186 respondents were female (54.2%). Satisfaction ($\alpha = 0.90$), perceived emotional anthropomorphism ($\alpha = 0.82$), and perceived autonomous anthropomorphism ($\alpha = 0.84$) were measured using the same 7-point Likert scales as in the previous studies.

7.2 | Data Analysis

Manipulation checks. The service chatbot in the high autonomous anthropomorphism condition was perceived as more autonomous than the one in the low autonomous anthropomorphism condition ($M_{\text{low autonomy}} = 4.61$, SD = 1.15, $M_{\text{high autonomy}} = 5.25$, SD = 1.01, $t(341) = -5.48$, $p < 0.001$). Furthermore, participants in the high emotional anthropomorphism condition perceived the service chatbot as more emotional than the chatbot in the low emotional anthropomorphism condition ($M_{\text{low emotionality}} = 3.26$, SD = 1.35, $M_{\text{high emotionality}} = 3.74$, SD = 1.40, $t(341) = -3.28$, $p < 0.01$). To assess dimensional separability, we conducted 2×2 ANOVAs with both manipulation check measures as dependent variables. For the emotional manipulation check, the emotional manipulation had a significant main effect ($F(1, 339) = 10.59$, $p = 0.001$, $\eta^2 = 0.03$), confirming that the emotional manipulation successfully increased perceived emotional anthropomorphism. The autonomous manipulation also had a smaller but significant effect on the emotional check ($F(1, 339) = 8.85$, $p = 0.003$, $\eta^2 = 0.03$), and no significant interaction was observed ($F(1, 339) = 0.50$, $p = 0.480$). For the autonomous manipulation check, the autonomous manipulation had a significant main effect ($F(1, 339) = 29.21$, $p < 0.001$, $\eta^2 = 0.08$), confirming its effectiveness. Critically, the emotional manipulation did not significantly affect perceived autonomous anthropomorphism ($F(1, 339) = 0.16$, $p = 0.688$, $\eta^2 = 0.00$), and no significant interaction was observed ($F(1, 339) = 1.05$, $p = 0.306$). These results show that the emotional manipulation was cleanly discriminant, affecting only its focal dimension, while the autonomous manipulation primarily affected its focal dimension but with some cross-dimensional influence on perceived emotionality.

Satisfaction. The results of an ANOVA revealed a main effect for autonomous anthropomorphism on satisfaction ($M_{\text{low autonomy}} = 4.97$, SD = 1.19, $M_{\text{high autonomy}} = 5.54$, SD = 0.99, $F(1, 339) = 23.0$, $p < 0.001$, $\eta^2 = 0.06$). Consistent with previous studies, the effect of emotional anthropomorphism on satisfaction was not significant ($M_{\text{low emotionality}} = 5.29$, SD = 1.14, $M_{\text{high emotionality}} = 5.24$, SD = 1.11, $F(1, 339) = 0.29$, $p = 0.59$). However, we observed a significant interaction effect on satisfaction ($F(1, 339) = 5.41$, $p = 0.02$, $\eta^2 = 0.02$). In particular, when the service chatbot did not express emotion, autonomous anthropomorphism led to higher satisfaction ($M_{\text{low autonomy}} = 4.87$, SD = 1.25, $M_{\text{high autonomy}} = 5.70$, SD = 0.83, $F(1, 177) = 27.5$, $p < 0.001$). However, when the service chatbot showed emotion, autonomous anthropomorphism had no significant effect on satisfaction ($M_{\text{low autonomy}} = 5.08$, SD = 1.11, $M_{\text{high autonomy}} = 5.37$, SD = 1.10, $F(1, 162) = 2.81$, $p = 0.10$). See Figure 3.

Measurement model. To examine whether perceived autonomous anthropomorphism, perceived emotional anthropomorphism, and satisfaction constitute empirically separable constructs, we conducted a confirmatory factor analysis on the eleven measurement items. The hypothesized three-factor model with correlated factors fit the data well ($\chi^2(41) = 103.19$, CFI = 0.971, TLI = 0.962, RMSEA = 0.067, SRMR = 0.048), with all standardized loadings significant ($\lambda = 0.63$ – 0.91) and composite reliabilities of 0.85, 0.83, and 0.90 for autonomous anthropomorphism, emotional anthropomorphism, and satisfaction, respectively. The three-factor model fit significantly better than every more parsimonious alternative: merging autonomous anthropomorphism and satisfaction ($\Delta\chi^2(2) = 65.42$, $p < 0.001$), autonomous and emotional anthropomorphism ($\Delta\chi^2(2) = 261.32$, $p < 0.001$), emotional anthropomorphism and satisfaction ($\Delta\chi^2(2) = 327.66$, $p < 0.001$), or all items onto a single factor ($\Delta\chi^2(3) = 348.96$, $p < 0.001$; see Table SA4 in Appendix III). We note that the latent correlation between autonomous anthropomorphism and satisfaction was high ($\phi = 0.91$), reflecting the strong downstream effect of perceived autonomy on satisfaction in this design and the fact that the Study 3 autonomy items partly index competent task performance; the revised measures introduced in Study 4, which target anthropomorphic attribution specifically, address this overlap directly.

7.3 | Discussion

Study 3 provided more evidence for the proposed effects of autonomous and emotional anthropomorphism of service chatbots. Autonomous anthropomorphism had a significant positive effect on satisfaction, whereas emotional anthropomorphism did not exhibit a significant main effect. More importantly, a significant interaction effect was observed: the positive effect of autonomous anthropomorphism on satisfaction was significant only when emotional anthropomorphism was low, suggesting that emotional cues may attenuate the competence-driven benefits of autonomy in utilitarian service contexts. This interaction is consistent with the theoretical framework developed in this paper. From the SCM perspective, emotional cues that enhance warmth perceptions may simultaneously weaken competence perceptions due to the

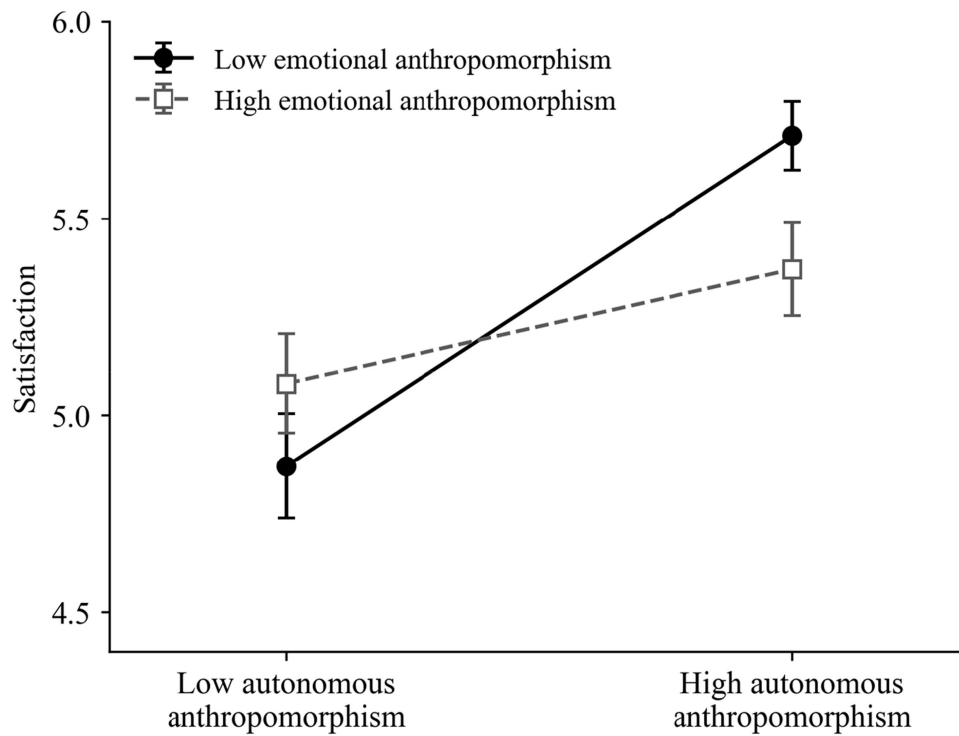


FIGURE 3 | The interaction effect of autonomous anthropomorphism and emotional anthropomorphism on satisfaction, Study 3.

compensatory relationship between these dimensions (Kervyn et al. 2010). From the EVT perspective, autonomous anthropomorphism likely constitutes a positive expectancy violation in utilitarian service contexts, whereas emotional expressions may be perceived as inauthentic or unnecessary, creating an ambiguous expectancy violation that reduces the positive effect of autonomy (Burgoon 2016). We note that, in the absence of a fully non-anthropomorphic baseline in Study 3, the observed pattern is also consistent with a conditional benefit of autonomous cues when emotional cues are absent, rather than necessarily a direct impediment from emotional cues. To disentangle these interpretations, and to further explore if such interaction happens in the translation of anthropomorphic cues into competence and warmth perceptions, we conducted a Study 4 with a non-anthropomorphic control condition, and with perceived competence and warmth measured.

8 | Study 4: The Interaction Effect Revisited: Nonanthropomorphic Baseline, Measured Mechanisms, and Revised Measures

8.1 | Research Method

A pre-registered Study 4 was designed to resolve the limitations of Study 3 and to provide the measurement evidence on which the central claims of this research rest. It replicates the 2 (autonomous anthropomorphism: high vs. low) $\times$ 2 (emotional anthropomorphism: high vs. low) between-subjects design of Study 3 with three critical modifications. First, the low-autonomy/low-emotion cell was constructed as a fully non-anthropomorphic baseline: the chatbot displayed no human name, no avatar, and no first-person language, and responded with strictly functional, template-like messages. This baseline allows the interaction to be interpreted against a true reference

point, distinguishing whether emotional cues actively attenuate the benefits of autonomy or whether autonomous cues merely confer benefits in the absence of emotional cues. Second, perceived warmth and perceived competence were measured as mediators, together with perceived helpfulness as a measured control, so that the evaluative mechanism behind the interaction could be demonstrated rather than assumed, and so that a moderated mediation analysis could adjudicate between a first-stage account (emotional cues dilute the translation of autonomous cues into competence perceptions) and a second-stage account (emotional cues change the weights that competence and warmth perceptions carry for satisfaction). Third, all three perceived anthropomorphism dimensions were measured with revised multi-item scales designed to capture anthropomorphic attributions specifically, enabling direct factor-analytic tests of the separability of the three dimensions from each other and from warmth and competence.

In the experiment, participants read a flight-booking scenario in which a user chatted with the TripPlus customer-service chatbot to book a flight to Shanghai, and then viewed a screenshot of the conversation corresponding to their condition (see Appendix SIV for the stimuli). Physical anthropomorphic cues were held constant at zero across all four conditions: no condition displayed a human name, human avatar, or other identity cues, so that the manipulations varied only autonomous and emotional cues. In the high-autonomy conditions, the chatbot proactively searched for alternatives, offered suggestions without explicit requests, and adapted its recommendations to the user's stated preferences; in the low-autonomy conditions, it provided rigid, reactive responses that answered only what was asked. In the high-emotion conditions, the chatbot's responses included empathetic language and emojis conveying enthusiasm and care; in the low-emotion conditions,

responses were strictly factual and devoid of emotional expression. As in the previous studies, all conditions presented identical flight information and led to the same service outcome within a comparable number of conversational turns.

We recruited 401 participants from Prolific. Two factual attention checks were included, asking participants to identify the destination city in the conversation and the type of service agent shown. Twenty-two participants (5.5%) failed at least one attention check and were excluded, leaving a final sample of 379 (cell sizes 92 to 97). The average age was 32.1 years ($SD = 10.6$), and 40.4% of the participants were female ($n = 153$). In terms of education, 11.9% had completed high school or below, 12.4% held a college or vocational degree, 62.5% held a bachelor's degree, and 13.2% held a master's degree or above; 98.4% reported prior experience interacting with chatbots.

All constructs were measured on 7-point scales (1 = *strongly disagree*, 7 = *strongly agree*). Perceived autonomous anthropomorphism was measured with four items capturing attributions of independent thinking and decision-making, drawing on the agency facet of mind perception (Gray et al. 2007; Malle 2019) and the “think” dimension of brand anthropomorphism (Huang et al. 2020): “The chatbot seems able to think,” “The chatbot seems able to make decisions on its own,” “The chatbot anticipates what the user needs,” and “The chatbot seems capable of planning” ($\alpha = 0.89$). Perceived emotional anthropomorphism was measured with three items targeting the attribution of emotional capacity rather than the intensity of emotional displays, drawing on the experience facet of mind perception (Gray et al. 2007; Malle 2019) and the “feel” dimension (Huang et al. 2020): “The chatbot seems able to experience emotions,” “The chatbot seems to have feelings of its own,” and “The chatbot seems able to sense the user's emotions” ($\alpha = 0.94$). Perceived physical anthropomorphism was measured with four items covering both appearance and identity cues, consistent with the definition of the dimension (Seeger et al. 2021; the appearance items were adapted from Belanche et al. 2021): “The chatbot's appearance looks human,” “The chatbot's appearance looks like a machine” (reverse-coded), “The chatbot has human identity cues (e.g., a human name and avatar),” and “The chatbot looks like it could plausibly be a real person” ($\alpha = 0.80$). Perceived competence (competent, intelligent, efficient, skillful, capable; $\alpha = 0.95$) and perceived warmth (friendly, warm, good-natured, sincere; $\alpha = 0.91$) were measured with the trait sets of the stereotype content model (Fiske et al. 2002). Perceived helpfulness was measured with two items (“The chatbot is helpful to the user,” “The service the chatbot provides is useful”; $\alpha = 0.92$, $r = 0.85$). Customer satisfaction was measured with three items ($\alpha = 0.92$) (Brady et al. 2005). The full item wordings and sources are reported in Table SA5 in Appendix IV. All experimental hypotheses, 2×2 factorial design, participant sampling criteria, manipulation protocols, mediation analytical plan, and exclusion rules were pre-registered prior to data collection on AsPredicted (Preregistration link: <https://aspredicted.org/jz6er7.pdf>). No unplanned confirmatory analyses were conducted. No deviations from the pre-registered analysis plan occurred.

8.2 | Data Analysis

Manipulation checks. The manipulations were effective and, with the revised measures, substantially cleaner than in the previous studies. A 2×2 ANOVA on perceived autonomous anthropomorphism revealed only a main effect of the autonomous manipulation ($M_{\text{low autonomy}} = 4.89$, $SD = 1.29$ vs. $M_{\text{high autonomy}} = 5.69$, $SD = 1.02$; $F(1, 375) = 44.34$, $p < 0.001$, $\eta^2 = 0.106$); neither the emotional manipulation ($F(1, 375) = 0.03$, $p = 0.853$) nor the interaction ($F(1, 375) = 0.02$, $p = 0.885$) had any effect, indicating that the autonomous manipulation and measure were fully discriminant. A parallel ANOVA on perceived emotional anthropomorphism showed a strong main effect of the emotional manipulation ($M_{\text{low emotionality}} = 2.89$, $SD = 1.51$ vs. $M_{\text{high emotionality}} = 4.02$, $SD = 1.68$; $F(1, 375) = 48.39$, $p < 0.001$, $\eta^2 = 0.114$), a much smaller cross-over effect of the autonomous manipulation ($F(1, 375) = 7.68$, $p = 0.006$, $\eta^2 = 0.020$), and no interaction ($F(1, 375) = 0.07$, $p = 0.794$). As in Studies 2 and 3, a proactive, adaptive chatbot was perceived as somewhat more emotionally engaged, but the effect of the focal manipulation was more than five times larger. Finally, perceived physical anthropomorphism, which was measured but not manipulated, remained low and undifferentiated across conditions (cell means 2.24 to 2.63; autonomous manipulation $F(1, 375) = 1.77$, $p = 0.185$; emotional manipulation $F(1, 375) = 1.66$, $p = 0.199$; interaction $F(1, 375) = 3.65$, $p = 0.057$), confirming that the autonomous and emotional manipulations did not inadvertently move perceptions of the physical dimension and that all conditions were free of physical anthropomorphic cues. Table 4 reports the means and standard deviations of all measured variables by condition.

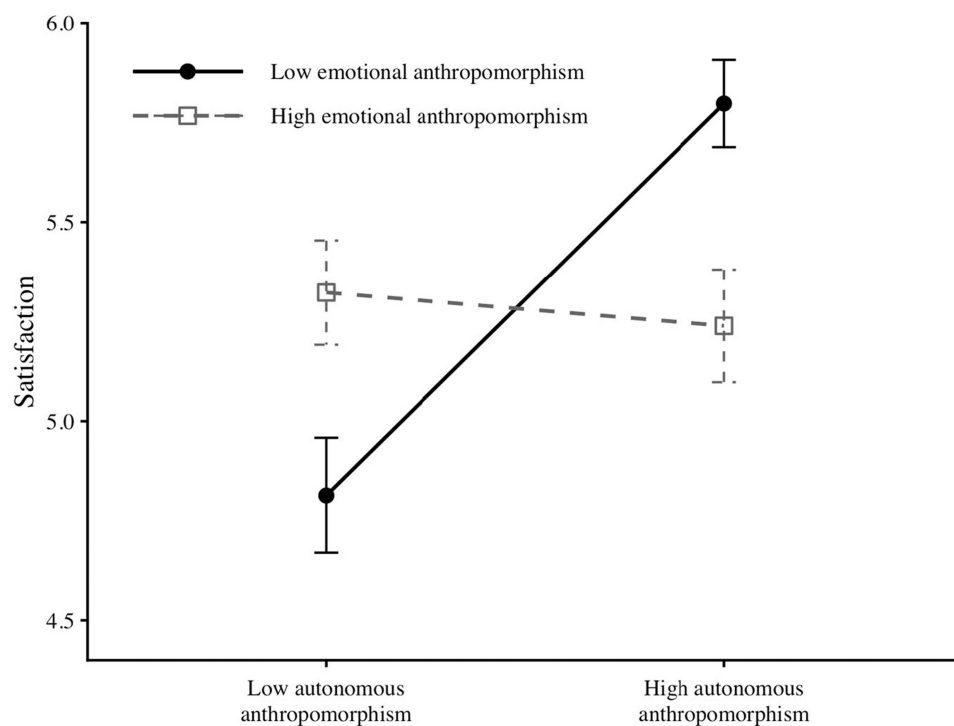
Satisfaction. A 2×2 ANOVA on satisfaction revealed a main effect of autonomous anthropomorphism ($F(1, 375) = 11.98$, $p < 0.001$, $\eta^2 = 0.031$), no main effect of emotional anthropomorphism ($F(1, 375) = 0.03$, $p = 0.861$), and, critically, a significant interaction ($F(1, 375) = 16.33$, $p < 0.001$, $\eta^2 = 0.042$), replicating the interaction observed in Study 3 (see Figure 4). Simple-effects analyses showed that when emotional anthropomorphism was low, autonomous anthropomorphism strongly increased satisfaction ($M_{\text{low autonomy}} = 4.81$, $SD = 1.41$ vs. $M_{\text{high autonomy}} = 5.80$, $SD = 1.08$; $F(1, 375) = 28.11$, $p < 0.001$), whereas when emotional anthropomorphism was high, it had no effect ($M = 5.32$, $SD = 1.28$ vs. $M = 5.24$, $SD = 1.36$; $F(1, 375) = 0.20$, $p = 0.656$), supporting H5.

The inclusion of the fully non-anthropomorphic baseline permits the finer-grained reading that Study 3 could not provide. Relative to the baseline ($M = 4.81$), each type of anthropomorphic cue was beneficial on its own: the autonomy-only cell was substantially higher ($\Delta M = 0.98$, $t(375) = 5.30$, $p < 0.001$), and the emotion-only cell was also higher ($\Delta M = 0.51$, $t(375) = 2.73$, $p = 0.007$). The combined cell exceeded the baseline ($\Delta M = 0.43$, $t(375) = 2.26$, $p = 0.024$) but was significantly lower than the autonomy-only cell ($\Delta M = -0.56$, $t(375) = -2.99$, $p = 0.003$) and no different from the emotion-only cell ($t(375) = -0.45$, $p = 0.656$). Framed as simple effects of the emotional manipulation, emotional cues increased satisfaction when added to a non-autonomous chatbot ($F(1, 375) = 7.45$, $p = 0.007$) but decreased satisfaction when added to an autonomous chatbot ($F(1, 375) = 8.91$, $p = 0.003$). This crossover pattern resolves the interpretive ambiguity of Study 3:

TABLE 4 | Means and standard deviations of all measured variables by condition, Study 4.

Measure	Low A/low E (baseline) (n = 95)	Low A/high E (n = 95)	High A/low E (n = 97)	High A/high E (n = 92)
Perceived autonomous anthropomorphism	4.87 (1.35)	4.91 (1.22)	5.69 (0.94)	5.69 (1.11)
Perceived emotional anthropomorphism	2.65 (1.32)	3.82 (1.58)	3.14 (1.64)	4.22 (1.75)
Perceived physical anthropomorphism	2.24 (1.16)	2.62 (1.15)	2.63 (1.17)	2.55 (1.14)
Perceived competence	4.81 (1.24)	5.21 (1.17)	5.78 (1.01)	5.17 (1.20)
Perceived warmth	4.43 (1.37)	4.77 (1.23)	4.44 (1.35)	4.76 (1.31)
Perceived helpfulness	5.86 (1.11)	5.76 (0.96)	6.18 (0.92)	6.15 (0.95)
Customer satisfaction	4.81 (1.41)	5.32 (1.28)	5.80 (1.08)	5.24 (1.36)

Note: A = autonomous anthropomorphism manipulation; E = emotional anthropomorphism manipulation. Standard deviations in parentheses. All measures on 7-point scales.

**FIGURE 4** | The interaction effect of autonomous anthropomorphism and emotional anthropomorphism on satisfaction, Study 4.

the data support the stronger reading that emotional cues actively undermine the benefits of autonomy, rather than the weaker reading that autonomous cues merely happen to help only when emotional cues are absent. Emotional expressions are not detrimental per se; they become detrimental specifically in combination with autonomous cues.

Perceived competence and warmth. The interaction on satisfaction was mirrored precisely, and exclusively, in perceived competence. A 2×2 ANOVA on competence yielded a main effect of autonomy ($F(1, 375) = 15.90, p < 0.001, \eta^2 = 0.041$) qualified by a significant interaction ($F(1, 375) = 18.43, p < 0.001, \eta^2 = 0.047$): autonomy increased competence when emotional cues were absent ($M = 4.81, SD = 1.24$ vs. $M = 5.78, SD = 1.01; F(1, 375) = 34.26, p < 0.001$) but not when they were

present ($M = 5.21, SD = 1.17$ vs. $M = 5.17, SD = 1.20; F(1, 375) = 0.07, p = 0.799$), and adding emotional cues to an autonomous chatbot significantly reduced perceived competence ($\Delta M = -0.62, F(1, 375) = 13.50, p < 0.001$). Perceived warmth, by contrast, showed only a modest main effect of the emotional manipulation ($M_{\text{low emotionality}} = 4.43$ vs. $M_{\text{high emotionality}} = 4.77; F(1, 375) = 5.91, p = 0.016, \eta^2 = 0.016$), with no effect of autonomy and no interaction (both F s < 0.01). To test formally whether the interaction was specific to the competence pathway, we computed a $2 \times 2 \times 2$ mixed ANOVA with perception type (competence vs. warmth) as a within-subjects factor; the three-way interaction was significant ($F(1, 375) = 15.90, p < 0.001, \eta^2 = 0.041$), confirming that the autonomy-by-emotion interaction operated on competence perceptions and not on warmth perceptions.

Ruling out perceived helpfulness. Perceived helpfulness showed a main effect of autonomy ($F(1, 375) = 12.24$, $p < 0.001$, $\eta^2 = 0.031$) but, importantly, no interaction with the emotional manipulation ($F(1, 375) = 0.11$, $p = 0.744$). The dissociation is informative: if the satisfaction effects reflected instrumental usefulness rather than anthropomorphic inference, the interaction pattern should have appeared on helpfulness as well. Moreover, when perceived helpfulness was included as a covariate, the autonomy-by-emotion interaction on satisfaction remained significant and indeed grew stronger ($F(1, 374) = 39.11$, $p < 0.001$, $\eta^2 = 0.095$), as did the interaction on perceived competence ($F(1, 374) = 45.70$, $p < 0.001$, $\eta^2 = 0.109$). The effects of autonomous anthropomorphism and its attenuation under emotional cues thus hold over and above perceived usefulness. The key results were also robust to controlling for age and gender (interaction on satisfaction: $F(1, 373) = 15.63$, $p < 0.001$).

Moderated mediation. To test the mechanism behind the interaction, we estimated a first-stage moderated mediation model with the autonomous manipulation as the independent variable, the emotional manipulation as the moderator of the paths to the mediators, perceived competence and perceived warmth as parallel mediators, and satisfaction as the outcome (10,000 bootstrap samples). In the mediator model, the autonomous manipulation increased perceived competence ($b = 0.98$, $p < 0.001$), and this effect was significantly reduced under emotional cues (autonomy $\times$ emotion $b = -1.02$, $p < 0.001$); perceived warmth was unaffected by the autonomous manipulation ($b = 0.004$, $p = 0.983$) and showed no interaction ($b = -0.01$, $p = 0.982$). In the outcome model, perceived competence ($b = 0.71$, $p < 0.001$) and perceived warmth ($b = 0.21$, $p < 0.001$) both positively predicted satisfaction, supporting H4a and H4b. The conditional indirect effect of autonomous anthropomorphism through perceived competence was significant when emotional anthropomorphism was low (0.69, 95% CI: [0.45, 0.93]) but not when it was high (-0.03 , 95% CI: $[-0.28, 0.21]$), and the index of moderated mediation was significantly negative (-0.72 , 95% CI: $[-1.07, -0.38]$). The indirect effects through warmth were null at both levels of the moderator (index = -0.001 , 95% CI: $[-0.12, 0.12]$). Consistent with H3b, the satisfaction effect of autonomous anthropomorphism, and its attenuation under emotional cues, was transmitted through

perceived competence and not through perceived warmth. The pattern was robust to including perceived helpfulness as a covariate (index of moderated mediation = -0.45 , 95% CI: $[-0.65, -0.28]$); with helpfulness controlled, the conditional indirect effect through competence was significantly negative under high emotional cues (-0.16 , 95% CI: $[-0.28, -0.06]$), indicating that emotional cues did not merely neutralize but reversed the competence pathway once instrumental value was held constant. Table 5 summarizes the conditional indirect effects.

We then compared this first-stage account against the second-stage alternative, under which emotional cues change the weights that the perceptions carry for satisfaction. Augmenting the outcome model with perception-by-moderator terms revealed that second-stage moderation was also present: under emotional cues, the weight of competence for satisfaction decreased ($b = -0.28$, 95% CI: $[-0.45, -0.12]$) and the weight of warmth increased ($b = 0.29$, 95% CI: $[0.13, 0.45]$). The second-stage index of moderated mediation through competence (-0.28 , 95% CI $[-0.48, -0.11]$) was, however, substantially smaller than the first-stage index (-0.72), and in the full dual-stage model the conditional indirect effect through competence remained significant at low emotion (0.80, 95% CI: $[0.53, 1.10]$) and null at high emotion (-0.02 , 95% CI: $[-0.21, 0.15]$). The data thus support the first-stage account as the primary locus of the moderation, as our framework predicts: the co-presence of emotional cues chiefly dilutes the translation of autonomous cues into competence perceptions. At the same time, the second-stage evidence that emotional cues shift evaluative weight from competence toward warmth is theoretically coherent with the SCM logic and suggests that mixed cue combinations alter not only what users perceive but also which perceptions they consult, a nuance we return to in the general discussion.

Measurement model: factor structure and discriminant validity. Study 4's item-level data permit a direct test of the empirical separability of the three anthropomorphism dimensions. A confirmatory factor analysis of the eleven perceived-anthropomorphism items specified as three correlated factors fit the data well ($\chi^2(41) = 106.51$, CFI = 0.976, TLI = 0.968, RMSEA = 0.065, SRMR = 0.043). All alternative

TABLE 5 | Conditional indirect effects of autonomous anthropomorphism on satisfaction, Study 4.

Pathway	Emotional anthropomorphism	Indirect effect	Boot SE	95% CI
Through perceived competence	Low	0.69	0.12	[0.45, 0.93]
	High	-0.03	0.12	$[-0.28, 0.21]$
Index of moderated mediation		-0.72	0.18	$[-1.07, -0.38]$
Through perceived warmth	Low	0.001	0.04	$[-0.09, 0.09]$
	High	-0.0004	0.04	$[-0.08, 0.08]$
Index of moderated mediation		-0.001	0.06	$[-0.12, 0.12]$
With helpfulness as covariate				
Through perceived competence	Low	0.29	0.06	[0.18, 0.42]
	High	-0.16	0.06	$[-0.28, -0.06]$
Index of moderated mediation		-0.45	0.09	$[-0.65, -0.28]$

Note: First-stage moderated mediation with parallel mediators; 10,000 bootstrap samples; percentile confidence intervals.

structures fit substantially worse: a one-factor model treating anthropomorphism as unitary ($\chi^2(44) = 1175.33$, CFI = 0.592, RMSEA = 0.261; $\Delta\chi^2(3) = 1068.82$, $p < 0.001$), a two-factor model merging the two mental dimensions into a single mind factor alongside the physical factor ($\chi^2(43) = 1138.21$, CFI = 0.605; $\Delta\chi^2(2) = 1031.70$, $p < 0.001$), and the best-fitting two-factor alternative, which merged the emotional and physical dimensions ($\chi^2(43) = 431.55$, CFI = 0.860, RMSEA = 0.155; $\Delta\chi^2(2) = 325.04$, $p < 0.001$). The three perceived dimensions are thus positively correlated, as expected for facets of a higher-order construct, but empirically separable.

To address the concern that the anthropomorphism measures may capture adjacent constructs, we estimated a full measurement model adding perceived warmth and perceived competence, specifying five correlated factors across twenty items. This model fit well ($\chi^2(160) = 477.27$, CFI = 0.949, TLI = 0.940, RMSEA = 0.072, SRMR = 0.052), and constraining the theoretically most worrisome construct pairs to single factors substantially degraded fit: merging autonomous anthropomorphism with competence ($\Delta\chi^2(4) = 481.98$, $p < 0.001$, CFI = 0.873) and merging emotional anthropomorphism with warmth ($\Delta\chi^2(4) = 943.53$, $p < 0.001$, CFI = 0.799). Composite reliabilities ranged from 0.81 to 0.95, and the average variance extracted exceeded 0.50 for every construct (autonomous = 0.68, emotional = 0.84, physical = 0.54, competence = 0.79, warmth = 0.71). The Fornell-Larcker criterion was satisfied for every construct pair; the largest latent correlation, between autonomous anthropomorphism and competence ($r = 0.69$), remained below the square root of the AVE of both constructs (0.82 and 0.89). Likewise, all HTMT ratios were below the conservative 0.85 threshold (maximum = 0.70, again for autonomous anthropomorphism and competence). Perceiving a chatbot as capable of thinking, planning, and deciding on its own is thus related to, but empirically distinct from, judging it competent, and attributing emotional capacity to it is distinct from finding it warm. Table 6 reports the model comparisons; the full inter-construct correlation

matrix and HTMT ratios are reported in Web Appendix Table SA6.

8.3 | Discussion

Study 4 accomplishes three things. First, it replicates the central interaction of Study 3 with a cleaner design and traces its mechanism. The benefit of autonomous anthropomorphism for satisfaction, present and substantial when emotional cues were absent, disappeared when emotional cues were added, and moderated mediation analyses located this attenuation where our framework predicts: in the translation of autonomous cues into competence perceptions. Emotional cues significantly reduced the perceived competence of an otherwise identical autonomous chatbot, and it was this diluted competence perception, not any change in warmth, that carried the interaction into satisfaction. Second, the fully non-anthropomorphic baseline resolves the interpretive ambiguity that limited Study 3. Because emotional cues alone improved satisfaction relative to the baseline while significantly reducing satisfaction when combined with autonomous cues, the pattern constitutes a genuine cross-over: emotional cues actively undermine the benefits of autonomy rather than autonomy merely benefiting from their absence. Under the mixed-signals logic of the SCM and the expectancy-violation account developed earlier, an agent that simultaneously signals independent capability and displays emotional expressions presents the user with conflicting evidence about what kind of agent it is, and the competence inference that autonomous behavior would otherwise license is discounted. Third, Study 4 supplies the measurement evidence on which the multi-dimensional framework rests. With revised multi-item measures, the three perceived dimensions formed the hypothesized three-factor structure, outperformed all one- and two-factor alternatives, and showed discriminant validity against warmth and competence, including for the construct pair of greatest concern, autonomous anthropomorphism and competence.

Three further aspects of the results warrant comment. First, unlike in Study 2, the autonomous manipulation did not elevate

TABLE 6 | Confirmatory factor analysis model comparisons, Study 4.

Model	χ^2	df	CFI	TLI	RMSEA	SRMR	$\Delta\chi^2$ (Δdf) vs. hypothesized
Perceived anthropomorphism (11 items)							
Three-factor (autonomous, emotional, physical)	106.51	41	0.976	0.968	0.065	0.043	—
Two-factor (emotional + physical merged)	431.55	43	0.860	0.821	0.155		325.04 (2)***
Two-factor (autonomous + emotional merged)	1138.21	43	0.605	0.495	0.260		1031.70 (2)***
One-factor	1175.33	44	0.592	0.490	0.261		1068.82 (3)***
Full measurement model (20 items)							
Five-factor (three dimensions + warmth + competence)	477.27	160	0.949	0.940	0.072	0.052	—
Four-factor (autonomous + competence merged)	959.25	164	0.873	0.852	0.113		481.98 (4)***
Four-factor (emotional + warmth merged)	1420.80	164	0.799	0.767	0.142		943.53 (4)***
One-factor	3340.72	170	0.492	0.432	0.222		2863.45 (10)***

Note: $N = 379$. *** $p < 0.001$. The two-factor model merging the autonomous and physical dimensions ($\chi^2(43) = 713.54$, CFI = 0.758) also fit substantially worse than the three-factor model.

perceived warmth. We attribute this to the sharpened Study 4 measures, which target the attribution of emotional capacity and social warmth rather than general engagement, and to the sharpened manipulations; with cleaner instruments, each manipulation moved only its theoretically focal perception, which is precisely the dimension-specific pattern the framework predicts. Second, warmth was not evaluatively inert: perceived warmth positively predicted satisfaction (supporting H4b), and under emotional cues its evaluative weight increased while that of competence decreased. Warmth's null role in the indirect effects reflects the fact that the manipulations barely moved it, not that it carries no weight. This refines the interpretation of Study 2: in utilitarian contexts, competence is the dominant, but not the only, evaluative pathway. Third, relative to a truly nonanthropomorphic chatbot, every anthropomorphic configuration improved satisfaction, including emotional cues alone. The managerial implication is therefore not that emotional design is harmful, but that anthropomorphic cues are not additive: emotional expressions helped a basic chatbot yet erased the considerably larger gains of an autonomous one. Designers face a choice between cue configurations, and in utilitarian contexts the autonomy-only configuration dominated all others.

8.4 | Implications and Conclusion

This research proposed and tested a three-dimensional framework of service chatbot anthropomorphism—comprising physical, emotional, and autonomous anthropomorphism—and examined how these dimensions independently and interactively influence customer satisfaction through the evaluative mechanisms of perceived warmth and competence. Across four studies using a mixed-method approach, we demonstrate that the effects of anthropomorphism on satisfaction depend critically on which dimension is activated, challenging the prevailing treatment of anthropomorphism as a unitary construct.

8.5 | Theoretical Contributions

This research makes three primary contributions to the anthropomorphism and human-computer interaction literatures. First, by identifying physical, emotional, and autonomous anthropomorphism as distinct dimensions grounded in users' psychological attributions, this research offers a more integrative framework than prior taxonomies organized around design modality (e.g., Seeger et al. 2021) or limited to two dimensions (e.g., Huang et al. 2020). The framework provides a structured lens through which previously mixed findings can be better understood. For example, the inconsistency between Araujo (2018) finding that anthropomorphic cues increased social presence but not attitudes and Yuan and Dennis's (2019) finding that visual but not auditory cues increased willingness to pay can be partly explained by the fact that these studies activated different anthropomorphism dimensions with different evaluative consequences.

Second, by integrating the stereotype content model into the study of chatbot anthropomorphism, this research specifies the evaluative mechanisms through which each dimension

operates. We demonstrate that autonomous anthropomorphism uniquely enhances satisfaction through perceived competence, whereas physical and emotional anthropomorphism primarily influence perceived warmth. In utilitarian service contexts, where functional goals drive the interaction, competence perceptions are more consequential for satisfaction than warmth perceptions (Liu et al. 2022). This finding extends prior work on warmth and competence in human-robot interaction (Belanche et al. 2021; Reeves et al. 2020) by showing that the relative importance of these evaluative dimensions depends on the service context and the specific anthropomorphism dimension activated.

Third, this research is among the first to examine the interactive effects of different anthropomorphism dimensions. The finding that the positive effect of autonomous anthropomorphism on satisfaction was attenuated by emotional anthropomorphism provides empirical evidence for the compensatory dynamics predicted by the SCM (Kervyn et al. 2010) and the differential expectancy violations predicted by EVT (Burgoon 2016) in the chatbot context. This interaction challenges the common assumption in both research and practice that adding more human-like features is always beneficial and highlights the importance of considering dimensional interplay when designing service chatbots.

8.6 | Understanding the Null Effects

An important pattern in our findings is that neither physical nor emotional anthropomorphism significantly affected customer satisfaction. This pattern deserves careful consideration, and the two dimensions warrant separate treatment because they failed to influence satisfaction for different reasons.

Emotional anthropomorphism behaved as the framework predicts in a utilitarian context. It significantly enhanced perceived warmth, supporting H2, yet this warmth did not translate into satisfaction; the indirect effect through warmth was not significant. One explanation is that in utilitarian service contexts, where the primary goal of the interaction is task completion, warmth perceptions are simply less diagnostic of service quality than competence perceptions. Users interacting with a service chatbot to book a flight or inquire about a product are primarily concerned with whether the chatbot can solve their problem, not with whether it is friendly or emotionally expressive. This interpretation is consistent with Liu et al.'s (2022) finding that competence is more predictive of favorable outcomes in utilitarian settings, whereas warmth becomes more important in hedonic contexts. The weakness of the warmth-to-satisfaction pathway in this context does not imply that warmth is irrelevant to chatbot evaluation; future research should examine whether this pattern reverses in hedonic or relational service contexts, where emotional connection is valued. Notably, Study 4 provides initial evidence that this pathway can reach significance with strengthened manipulations, although the effect size remains substantially smaller than the competence pathway.

Physical anthropomorphism, by contrast, failed at an earlier stage. The manipulation produced a directionally positive but nonsignificant effect on warmth, so H1 was not supported, and

there was accordingly no pathway through which it could influence satisfaction. We attribute this primarily to the weakness of the manipulation rather than to the irrelevance of the dimension. Name-and-avatar cues in a static, text-based screenshot may be too subtle to shift warmth perceptions reliably, whereas the spontaneous emergence of the physical dimension in consumers' own descriptions (Study 1) indicates that it is a meaningful category of anthropomorphic attribution.

It is also worth noting that the null effect of emotional anthropomorphism on satisfaction does not imply that emotional cues are without consequence. As Study 3 demonstrates, emotional anthropomorphism moderated the effect of autonomous anthropomorphism, suggesting that emotional cues play an important role in shaping the overall evaluation, albeit not as a direct driver of satisfaction in utilitarian contexts.

8.7 | Boundary Conditions

The findings of this research should be interpreted within the context of utilitarian service interactions. In this context, users approach the chatbot with specific functional goals, making competence perceptions particularly consequential for satisfaction. However, in other service contexts—such as healthcare (De Gennaro et al. 2020), education (Ahmad et al. 2019 February), or companionship applications—users may prioritize warmth, emotional support, and social presence over task efficiency. In such contexts, emotional anthropomorphism may play a more central role in driving satisfaction, and the compensatory tension between warmth and competence may operate differently.

Furthermore, the interaction between emotional and autonomous anthropomorphism observed in Study 3 and Study 4 may be moderated by individual differences. Users with a higher need for social connection (Epley et al. 2007) may respond more positively to emotional cues, even in utilitarian contexts, while users with a higher need for efficiency may be particularly sensitive to the diluting effect of emotional cues on competence perceptions. Cultural factors may also play a role, as expectations regarding emotional expression in service encounters vary across cultures. Future research should explore these potential moderators to establish the generalizability and boundary conditions of the interaction effect.

8.8 | Managerial Implications

Our findings offer actionable guidance for managers designing AI-driven service chatbots. The most important implication is that in utilitarian service contexts—such as banking, e-commerce, and travel booking—managers should prioritize autonomous capabilities over superficial human-like features. Chatbots that demonstrate adaptive problem-solving, contextual understanding, and proactive assistance are more likely to satisfy customers than those that simply display emotional warmth or human-like appearance.

Moreover, our findings caution against the common practice of layering emotional cues onto functionally capable chatbots.

Adding empathetic language, emojis, or emotional expressions to a chatbot that already demonstrates high autonomous capability may not enhance—and may even reduce—customer satisfaction. This suggests that chatbot designers should carefully consider which anthropomorphic features to include based on the service context and the primary needs of their users. That said, warmth-related features should not be dismissed entirely. Physical and emotional anthropomorphism contribute to warmth perceptions, which may be valuable in contexts where building rapport, trust, or long-term relationships with customers is the primary goal. Managers should therefore align their anthropomorphic design choices with the specific objectives of the chatbot interaction.

9 | Limitations and Future Research

This research focuses on service chatbots designed to satisfy users' utilitarian needs. However, chatbots are also widely used in hedonic contexts—such as education and healthcare—where users may expect greater warmth, emotional connection, and social presence (Ahmad et al. 2019, February; De Gennaro et al. 2020; Liu and Sundar 2018). In such contexts, relational qualities may play a more central role than efficiency. Consequently, the relative importance of anthropomorphic dimensions observed in this study may not generalize to nonutilitarian settings, where anthropomorphism may influence user satisfaction through different mechanisms.

Second, the experimental stimuli used in this research represent a simplified implementation of anthropomorphic cues. In Study 3 and Study 4, emotional and autonomous anthropomorphism were manipulated primarily through linguistic flexibility and emojis in a text-based chatbot interface. As conversational AI technologies continue to evolve—incorporating voice interaction, visual avatars, and more advanced forms of emotional expression—future research should examine how richer anthropomorphic cues influence user perceptions and evaluations. Relatedly, the relative importance of the physical dimension may differ in contexts where chatbots are embodied, such as service robots in restaurants or hotels, where physical appearance plays a more prominent role in shaping user perceptions.

Finally, the interaction examined in this research represents only one possible combination of anthropomorphism dimensions. Future research could investigate interactions between other dimensions, such as physical and emotional anthropomorphism or physical and autonomous anthropomorphism. In addition, scholars may extend the framework developed in this study to examine other outcomes—such as trust, moral evaluation, or responsibility attribution—building on prior work by Moreno and Terwiesch (2014) and Waytz et al. (2010).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: mar70255-sup-0001-Web_Appendix_PMAugust4.docx.